

Novel optimization approach for realized volatility forecast of stock price index based on deep reinforcement learning model

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ABSTRACT

Accurately predicting volatility has always been the focus of government decision-making departments, financial regulators and academia. Therefore, it is very crucial to precisely predict the realized volatility (RV) of the stock price index. In this paper, we take the RV sequences of Shanghai Stock Exchange Composite Index (SSEC), Standard & Poor 500 index (SPX) and Financial Times Stock Exchange Index (FTSE) as the research objects, and propose a predictive model based on optimized variational mode decomposition (VMD), deep learning models including deep belief network (DBN), long short-term memory network (LSTM) and gated recurrent unit (GRU), and reinforcement learning Q-learning algorithm. Firstly, the original RV sequence is decomposed by using the VMD ideal parameters optimized by grey wolf optimizer (GWO) to obtain the intrinsic mode functions (IMFs). Then, DBN, LSTM and GRU are used to predict same IMF simultaneously. Finally, the optimal weights of the above three models are determined by the Q-learning algorithm to construct an integrated model, and the final results are obtained after accumulating the predicted values of each IMF. The predictive performance of the model was evaluated by four loss functions: the mean average error (MAE), mean squared error (MSE), heterogeneous mean average error (HMAE), heterogeneous mean squared error (HMSE) and modified Diebold and Mariano test (MDM). The experimental results show that the constructed GVMD-Q-DBN-LSTM-GRU method has better performance than the comparison model in both emerging and developed markets.

1. Introduction

Volatility is commonly recognized as a measure of market uncertainty and an essential state variable for asset pricing and investment (Ghani, Guo, Ma, & Li, 2022; Liang, Wei, & Zhang, 2020). The volatility of stock price index will impact the investment behavior and returns of consumers, and even provide a significant shock to the stability of the financial environment (Mei, Liu, Ma, & Chen, 2017; Wang, Ma, Liu, & Yang, 2020). Improving the forecasting standard of stock price index volatility not only helps investors avoid risks and make successful profits, but also helps financial regulators to monitor the stock market and formulate macroeconomic management policies in a timely manner, as well as is important to promote the efficiency of financial markets and the stability of social order (Chun, Cho, & Ryu, 2020; Lu, Ma, Wang, Ding, & Wahab, 2021). Therefore, it is essential to establish an accurate stock volatility forecasting model.

To date, a variety of techniques have been proposed to solve volatility prediction task. According to a literature review by Poon and Granger (2003), most researches adopt classical stochastic volatility (SV) model (Taylor, 1994) and standard generalized autoregressive conditional heteroskedasticity (GARCH) model, which utilize the low-frequency trading data (Bollerslev, 1986; Engle, 1982).

To be specific, since the GARCH model can better describe the heteroskedasticity of the residuals of financial asset return series, a large number of scholars at home and abroad have extended the GARCH model, such as exponential GARCH (EGARCH) model (Nelson, 1991), nonlinear GARCH (NGARCH) model (Higgins, & Bera, 1992), asymmetric power GARCH (APGARCH) model (Ding, Granger, & Engle, 1993), threshold GARCH (TGARCH or GJR-GARCH) model (Zakoian, 1994), component GARCH (CGARCH) model (Hammoudeh, & Yuan, 2008), HM-EGARCH model (Lin, Xiao, & Li, 2020) and so on. Different from the conditional variance observable hypothesis in the GARCH

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model, the SV model considers that the conditional variance obeys an unobservable random process, which is more reasonable in setting of conditional variance (Chan, & Grant, 2016). Although SV is well able to model the time-varying variance data, it is still negatively affected by extreme outlier observations (Zhang, Li, Huang, Zhang, & Huang, 2022). Therefore, several extensions and improvements of SV model also exist for improving the prediction capability (Jacquier, Polson, & Rossi, 2004; Kastner, Fruhwirth-Schnatter, & Lopes, 2017; Schobel, & Zhu, 1999). It is worth affirming that the above GARCH-type models and SV model have achieved remarkable results, but only adopting low-frequency data such as daily data to characterize market volatility will lose a large amount of information on real market intraday trading price data, resulting in the high volatility state that the market may present on the day cannot be reflected (Luo, & Chen, 2020; Ma, Wei, Liu, & Huang, 2018).

In recent years, by the benefit of the rapid development in the ability to acquire and store high-frequency data, the application of high-frequency data has become a new important entry point for the study of financial asset volatility (Barndorff-Nielsen, & Shephard, 2004; Wen, Gong, & Cai, 2016). Andersen and Bollerslev (1998) defined RV on the basis of intraday high-frequency data (minute-level) as the sum of squares of intraday logarithmic returns, transforming the implied variable of financial asset volatility to be estimated into a directly observable explicit variable (McAleer, & Medeiros, 2008). On this basis, the modeling and prediction of RV has gradually emerged as a research hotspot due to its advantages of low noise and high measurement accuracy. Recent literature has documented that RV exhibits numerous common characteristics, the most typical of which is long memory (Andersen, Bollerslev, Diebold, & Labys, 2003; Ho, Shi, & Zhang, 2013; Takahashi, Watanabe, & Omori, 2021). For this reason, scholars are committed to constructing various discrete time series models to characterize the long memory of RV. For example, Hosking (1981) proposed autoregressive fractionally integrated moving average (ARFIMA) model, Barndorff-Nielsen (2002) proposed a flexible strategy to model the serial correlation of RV by superimposing the ARMA (1,1) process, Ghysels, Sinko and Valkanov (2007) used the method of mixed data sampling (MIDAS) to fit the long memory of RV. Corsi (2009) built a heterogeneous autoregressive (HAR) model for RV series through a combination of the autoregressive structure and the volatility components measured over different time ranges consistently outperforms the short memory model and is comparable to the more complex and cumbersome long-memory ARFIMA model. Luo, Chen and Wang (2023) constructed a time-varying HAR latent factor model with Bayesian method to forecast the RV of Chinese stock index futures and treasury bond futures, and the model outperformed the existing HAR models both in-sample and out-of-sample. Niu, Demirer, Suleman and Zhang (2023) incorporated the cross-sectional variance (CSV) component of asymmetric stock returns in the HAR model, which facilitates to promote the accuracy of RV forecasting. Although the underlying HAR model takes into account the heterogeneity of market participants, it fails to consider the jump in volatility of financial assets and leverage effect (Qiu, Zhang, Xie, & Zhao, 2019). In this regard, scholars have carried out a series extensions to the HAR model, with the typical ones being, Andersen, Bollerslev and Diebold (2007) considered the existence of jump in financial market to construct the HAR-RV-J and HAR-RV-CJ model with forecasts of RV for the foreign exchange spot market, the equity futures market and the interest rate futures market. Barndorff-Nielsen, Kinnebrock and Shephard (2008) decompose the RV of financial assets into the upstream RV (RS^+) and the downstream RV (RS^-) to construct the HAR-RS model. Combining with the heterogeneity of RV, leverage and jump rate, Corsi and Renò (2012) constructed the leverage heterogeneous autoregressive with continuous volatility and jumps (LHAR-CJ) model. Dutta and Das (2022) proposed to extend the HAR model using the jump information content of the implied volatility index (VIX) over time, and the presented HAR-RV method produces more precise forecasts than existing HAR-RV type models.

With the development of artificial intelligence (AI), the forecasting techniques represented by neural network have been widely applied in the early stage of time series modeling (Tang, de Almeida, & Fishwick, 1991). Artificial neural network (ANN) has the ability to learn the mapping relationship between the input and output sampled from the training set through supervised learning algorithms, thus it has been utilized to capture the volatility effect in stock returns with favorable forecasting results (Donaldson, & Kamstra, 1997; Kim, & Won, 2018). Gonzalez Miranda and Burgess (1997) predicted the implied volatility of Ibox35 index options via adopting a multilayer perceptron neural network, the empirical results show that it is capable of capturing the features of nonlinear relationship that contribute to volatility prediction. Nevertheless, the early neural networks have fallen into the low tide of research due to their susceptibility to overfitting and local extremum problem (Wang, Cui, Sun, & He, 2022). Hinton and Salakhutdinov (2006) proposed the DBN and corresponding efficient algorithms, which allow the entire neural network to generate training data according to the maximum probability through training the weights among neurons. This pioneered the research of deep learning (DL), and also became the dominant framework of DL in the future (Liu, Huang, Li, Chen, Liu, & Tai, 2023). DL models improve the comprehensiveness of the input by applying non-linear modules, which deepens the number of hidden layers based on neural network, and utilizes feature output of the low-level layer as high-level input to extract features from training data more efficiently (Ding, Zhang, Tao, & Li, 2022). Ribeiro, Santos, Mariani and Coelho (2021) predicted the RV of stocks by introducing the echo state network (ESN) model and HAR model at 1, 5 and 21 days ahead forecast periods. Recurrent neural network (RNN) can sufficiently connect the hidden neurons in the network, using the internal feedback mechanism to learn and maintain the memory of recent events (Chen, Wu, & Wu, 2022). However, RNN may face the problem of gradient disappearance or explosion, which is difficult to reflect long-term dependence (Pang, Zhou, Wang, Lin, & Chang, 2018). As a result, many scholars have modeled and carried out researches related to long memory (Huang, Tao, Liu, Sun, Yang, Gui, & Hu, 2022; Huang, Wei, Li, Yang, & Gui, 2022). In order to obtain both long-term and short-term information, Hochreiter and Schmidhuber (1997) established LSTM on the basis of RNN, which replaces the nodes of RNN with memory block that contain not only units capable of remembering temporal states, but also special gate structures to ensure more valuable and relevant information is filtered from training data (Liu, Yang, Huang, & Gui, 2020). Mucher (2022) extracted information from intraday high-frequency returns using LSTM model to predict the RV series of IBM stock. Petrozziello, Troiano, Serra, Jordanov, Storti, Tagliaferri and La Rocca (2022) investigated the profitability of LSTM to predict daily stock market fluctuations to support decision making in risk management applications. As another simpler and more efficient version of RNN, GRU has the advantages of reducing training time and improving network performance (Cho, Merriënboer, Gulcehre, Bahdanau, Bougares, Schwenk, & Bengio, 2014). Li, Zhou, Hu, Li, Su and Guo (2023) improved the prediction accuracy of RV by constructing the prediction model combining principal component analysis (PCA) and sparse principal component analysis (SPCA) methods with LSTM and GRU for the RV of two energy sector indices. In summary, DBN, LSTM and GRU have been successfully employed for financial time series forecasting owing to their ability to learn the intrinsic patterns in financial data and capture the nonlinear relationships in the data effectively (Lin, Yan, Xu, Liao, & Ma, 2021; Samarawickrama, & Fernando, 2017; Sezer, Gudelek, & Ozbayoglu, 2020; Zhang, Yan, & Aasma, 2020; Zheng, Fu, & Zhang, 2017).

As we all know, the single model is susceptible to random factors and has limitations and instability, and the integrated model constructed by utilizing the complementary advantages of multiple models can improve the overall generalization ability of the model (Peng, Zhang, Zhou, & Nazir, 2020). Some integrated forecasting models proposed in previous studies mostly use numerical methods and heuristic optimization

algorithms to improve the prediction effect of the integrated model (Liu, Yang, Wang, & Zhang, 2021; Liu, Yu, Wu, Duan, & Yan, 2020; Song, Wang, & Lu, 2018; Su, Liu, & Hong, 2019; Yu, Fan, Wang, Zhu, & Xu, 2022). Q-learning algorithm has received wide attention from scholars as reinforcement learning (RL) algorithm with excellent convergence and decision-making ability (Cheng, Zhao, Wang, Jiang, Xia, & Ding, 2016). It sets a reasonable reward and punishment mechanism according to the prediction error, and intelligently optimizes the weight coefficients adaptively through the characteristics of each forecasting model (Watkins, 1989).

Since RV exhibits typical volatility and irregularity affected by the influence of noise, preprocessing the original series prior to build the model contributes to facilitate more reliable forecasting (Li, Han, & Song, 2022). Data decomposition methods employed in the current stage of research are able to reduce the influence of high noise and randomness of time series on prediction, and further input information of different frequencies for the model (Huang, & Deng, 2021), such as, wavelet transform (WT) (Souropanis, & Vivian, 2020), empirical mode decomposition (EMD) (Huang et al., 1998), ensemble empirical mode decomposition (EEMD) (Zhang, Lin, & Shang, 2017), complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) (Lin, Lin, Liao, Li, Xu, & Yan, 2022). In comparison with the above decomposition methods, VMD is a novel adaptive non-recursive signal processing technique proposed by Dragomiretskiy and Zosso (2014), which has stronger robustness to measurement noise. In order to enhance the feature extraction effect of VMD, some scholars have used optimization algorithms such as genetic algorithm (GA), particle swarm optimization (PSO) to simultaneously search for the number of decomposed modes and quadratic penalty coefficient (Diao et al., 2020; Huang, Gao, Gan, & Ye, 2020; Li, Chen, Han, & Wang, 2020; Zhou, Yang, Shen, & Liu, 2020). Nonetheless, owing to the fact that GWO algorithm considers the leadership level and simulates the social level characteristics through a concise evolutionary scheme, it can significantly improve the overall performance and thus reduce the randomness of parameter selection effectively (Liu, Luo, Zhang, & Chen, 2021; Mirjalili, Mirjalili, & Lewis, 2014). In consequence, our paper utilizes the VMD method optimized by GWO algorithm to decompose RV.

Based on the above analysis and understanding, in this present study we propose a novel RV sequence prediction model GVMD-Q-DBN-LSTM-GRU, which consists of a decomposition method, deep learning and reinforcement learning techniques. First of all, our proposed model uses the VMD optimized by the GWO algorithm to decompose the RV sequence into IMFs of different frequencies. Next, the deep learning models DBN, LSTM and GRU are applied to make predictions for each IMF. Finally, the Q-learning algorithm is adopted to integrate the forecasting results of DBN, LSTM and GRU. The GVMD-Q-DBN-LSTM-GRU model is compared with other different models and its predictive performance was verified by four loss functions and MDM test (Diebold, & Mariano, 1995). The fundamental goal of this paper is to answer two questions: Does the GVMD-Q-DBN-LSTM-GRU model have the ability to accurately predict RV sequences? Is this approach equally predictive in emerging and developed markets?

The primary contributions of this article can be illustrated as follows:

(a) The VMD method optimized by GWO is used to preprocess the RV sequence of stock price index and decompose them into several subsequences, which fully reduces the nonlinear characteristics of the original RV sequence.

(b) Three deep neural networks are simultaneously applied to complete the prediction of same subsequences of RV sequence. In contrast to traditional statistical forecasting models and shallow neural networks, DBN, LSTM and GRU in deep neural networks possess powerful ability to deal with the high volatility of the original sequence due to the use of deep structure to explore the hidden features of RV sequences.

(c) The algorithm of reinforcement learning is employed for the integration of three prediction models. The reward and punishment mechanism of the reinforcement learning method manifests the

outstanding talent in adaptively optimizing the weight of the integrated model, and using the Early Stopping method to reasonably select the episodes during the training process, which addresses the problem of needing to manually set the number of episodes, thus making the value function Q converge stably. The combination of multiple deep learning models integrates the advantages of each model, which avoids the limitation that the traditional method is only applicable to a single condition, and facilitates the reliability and adaptability of the model.

(d) An integrated model composed of VMD method, three deep neural networks and Q-learning algorithm is proposed. The integrated deep reinforcement learning model is rarely used in RV sequence prediction. For this reason, the predictive efficiency of the combination of the two is a topic worth investigating. In order to verify the accuracy of the proposed GVMD-Q-DBN-LSTM-GRU model, several experiments are conducted under multiple sets of data to make comparisons with the competing models.

The rest of this paper is structured as follows. Section 2 introduces the basic methodology in this study and furthermore introduce the overall framework. Section 3 presents the data sources and experimental settings, and analyzes the results to confirm the efficiency and robustness of GVMD-Q-DBN-LSTM-GRU model. Section 4 provides the conclusion of the presented study and the prospect of future research.

2. Theoretical basis related to the combined forecasting model

In this subsection, the theoretical approaches of each method in the study are described in detail, including the adaptive data preprocessing technique, deep neural network forecasting approach, the Q-learning optimization model ensemble strategy and the prediction architecture of combined model.

2.1. Adaptive data preprocessing technique

In accordance with the methodology proposed by Andersen and Bollerslev (1998), we employ the intraday stock index returns to construct the daily RV. Statistically, for a given trading day t , $RV(t)$ can be calculated as the sum of the squares of the i -th ($i = 1, 2, \dots, M$) intraday return $r_{t,i}$:

$$RV(t) = \sum_{i=1}^M r_{t,i}^2 \quad (1)$$

$$r_{t,i} = 100 * (\ln P_{t,i} - \ln P_{t,i-1}) \quad (2)$$

where $1/M$ represents the sampling frequency, $P_{t,i}$ refers to the i -th closing price of the stock index on trading day t .

2.1.1. Grey wolf optimizer

The GWO algorithm is a novel meta-heuristic algorithm with a simple structure and few parameters, which simulates the predatory behavior and collaboration mechanisms arising from the social hierarchy driven by grey wolves in nature. During the implementation of the GWO algorithm, the initial population comprises is categorized into four grey wolf types alpha (α), beta (β), delta (δ) and omega (ω), which have their own responsibilities, thus naturally forming a hierarchical pyramid structure as presented in Fig. 1. As observed, the (α) wolves in the first stratum is the most dominant in the pack, and the combination is dominated and determined by a few number of (α) wolves (usually only one or two); The (β) wolves in the second stratum acts as followers of (α) wolves, complements the direction of (α) wolves in the cluster. The number of (β) wolves engaged in reconnaissance and hunting is more numerous than (α) wolves but less numerous than (δ) and (ω) wolves. Among the pack, (α) and (β) wolves are the most experienced. The role of (δ) wolves in the third stratum is responsible for assisting (α) and (β) wolves, but to dominate (ω) wolves. The (ω) wolves in the fourth stratum belongs to the lowest rank of grey wolves and is ruled by other wolves, yet an indispensable role in balancing the internal relationship

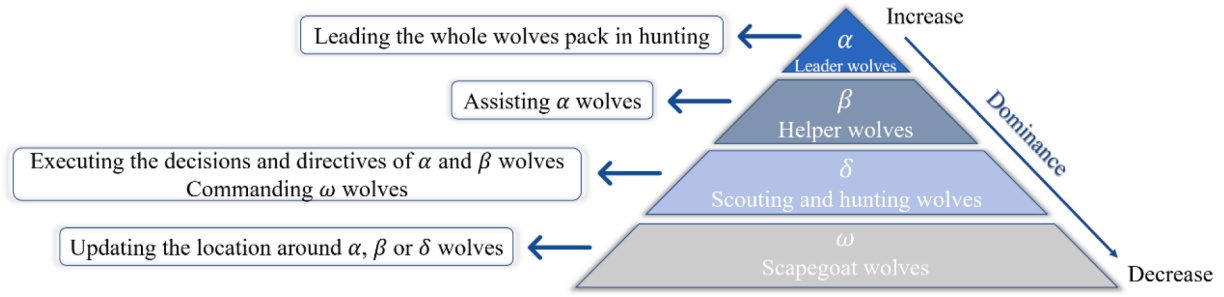


Fig. 1. Grey wolf hierarchy with a pyramid structure.

within the population.

In addition, several major stages of hunting (optimization), encircling the prey and attacking the prey and searching the prey are implemented as below.

(i) In the process of encircling the prey, the behavior of the grey wolves surrounding their prey is mathematically defined in Eqs. (3) and (4).

$$\vec{D} = \left| \vec{C} \bullet \vec{V}_{p,l} - \vec{V}_l \right| \quad (3)$$

$$\vec{V}_{l+1} = \left| \vec{V}_{p,l} - \vec{A} \bullet \vec{D} \right| \quad (4)$$

where $\vec{D}$ represents the distance between the closest wolf and prey, $\vec{V}_l$ and $\vec{V}_{p,l}$ represent the present positions of the wolf and prey at iteration number l , respectively. The swing factor $\vec{C}$ and convergence factor $\vec{A}$ are calculated as:

$$\vec{A} = 2\vec{a} \bullet \vec{r}_1 - \vec{a} \quad (5)$$

$$\vec{C} = 2 \bullet \vec{r}_2 \quad (6)$$

where $\vec{a}$ decreases linearly from 2 to 0 over each course of iteration, $\vec{r}_1$ and $\vec{r}_2$ are random vectors between $[0, 1]$.

(ii) In the process of grey wolf hunting, the GWO algorithm utilizes the position of the grey wolf to locate the prey, and guides other wolves to update their position and close to the prey in accordance with the best position of the individual grey wolf. The position of the prey is approximated as the average of (α) wolf, (β) wolf and (δ) wolf positions. The updated position of each wolf is described with the aid of Eqs. (7) to (9).

$$\begin{cases} \vec{D}_\alpha = \vec{C}_1 \bullet \vec{V}_\alpha - \vec{V}_t \\ \vec{D}_\beta = \vec{C}_2 \bullet \vec{V}_\beta - \vec{V}_t \\ \vec{D}_\delta = \vec{C}_3 \bullet \vec{V}_\delta - \vec{V}_t \end{cases} \quad (7)$$

$$\begin{cases} \vec{V}_1 = \vec{V}_\alpha - \vec{A}_1 \bullet \vec{D}_\alpha \\ \vec{V}_2 = \vec{V}_\beta - \vec{A}_2 \bullet \vec{D}_\beta \\ \vec{V}_3 = \vec{V}_\delta - \vec{A}_3 \bullet \vec{D}_\delta \end{cases} \quad (8)$$

$$\vec{V}_{l+1} = (\vec{V}_1 + \vec{V}_2 + \vec{V}_3) / 3 \quad (9)$$

where $\vec{D}_\alpha$, $\vec{D}_\beta$ and $\vec{D}_\delta$ separately denote the distances of the prey from the (α) wolf, (β) wolf and (δ) wolf, $\vec{V}_1$, $\vec{V}_2$ and $\vec{V}_3$ represent the positions of the (α) wolf, (β) wolf and (δ) wolf, respectively. $\vec{C}_1$, $\vec{C}_2$ and $\vec{C}_3$ separately indicate the swing factor of the (α) wolf, (β) wolf and (δ) wolf, $\vec{A}_1$, $\vec{A}_2$ and $\vec{A}_3$ indicate the convergence factor of the (α) wolf, (β) wolf

and (δ) wolf, respectively. $\vec{V}_{l+1}$ is the final position of the (ω) wolf.

(iii) As mentioned above, the process of attacking the prey by the grey wolf begins when the prey stops its movement. This procedure is done mathematically by decreasing the value of $\vec{a}$ from 2 to 0 over the course of iterations, which leads to a reduction in the fluctuation range of $\vec{A}$ as well. When the random value of $\vec{A}$ is in $[-1, 1]$, the (α) wolf is closer to the prey (and other wolves), however, when $A < -1$ or $A > 1$, the (α) wolf is further away from the prey (and other wolves), demonstrating that the GWO algorithm utilizes the attack mechanism of local search effectively.

(iv) It is worth mentioning that the process of searching is different from the process of attacking. During the searching stage, the wolves turn away from their companions in order to trace the prey ($A < -1$ or $A > 1$), and afterwards, the wolves approach each other in the attacking stage ($\vec{A}$ is in $[-1, 1]$). This process is called the divergence-convergence process in the phase of attacking.

As can be observed from Eq. (6), $\vec{C}$ contains random values in $[0, 2]$, which are considered as the obstacle of nature that slow down the speed of wolves to approach the prey. More specifically, depending on the location of the wolf, random weights are assigned to the prey in order to emphasize ($C > 1$) or attenuate ($C < 1$) the effect of the prey on the range in Eq. (3). In contrast to $\vec{A}$, $\vec{C}$ does not decrease linearly from 2 to 0, thus assuring the randomness of exploration in each iteration of the model.

2.1.2. Variational mode decomposition

Dragomiretskiy and Zosso (2014) proposed the VMD algorithm as an advanced decomposition technique for adaptive and non-recursive signal processing, which can reduce the effect of noise in financial data. By acquiring the optimum solution of the constrained variational model, the original input series $RV(t)$ is decomposed into k IMFs with optimal specific sparse characteristics, in which the sparse characteristics are measured by the spectral bandwidth of the series, that is, each IMF after VMD decomposition has the most compact bandwidth around its respective center frequency. The core idea of VMD method is to construct and solve the variational problem, and each step is described in detail below.

(i) Initially, in order to simplify the analysis of the signal, Hilbert transform is adopted to transform the real signal into an analytical signal, which is used to obtain the unilateral frequency spectrum of $u_k(t)$ ($k = 1, 2, \dots, K$), and realize the transformation of the signal from the bilateral frequency spectrum to the unilateral frequency spectrum. Let $f(t)$ be a real signal, whose Hilbert transform $\mathcal{H}[f(t)]$ is defined as follows.

$$\mathcal{H}[f(t)] = \hat{f}(t) = \frac{1}{\pi t} * f(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{f(\tau)}{t - \tau} d\tau = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{f(t - \tau)}{\tau} d\tau \quad (10)$$

It can be seen that the Hilbert transform of $f(t)$ is the convolution of $f(t)$ and $\frac{1}{\pi t}$, so the complex signal composed of the real signal $f(t)$ as the real part and its Hilbert transform $\hat{f}(t)$ as the imaginary part is called the

analytical signal, which is expressed in Eq. (11), the spectrum of $z(t)$ means the unilateral frequency spectrum.

$$z(t) = f(t) + j\hat{f}(t) \quad (11)$$

Subsequently, the spectrum of $u_k(t)$ is modulated to the corresponding fundamental frequency band through the exponential term $e^{-j\omega_k t}$ of the mixed center frequency ω_k . Finally, the squared L_2 norm of the modulated signal gradient is calculated to estimate the bandwidth of $u_k(t)$, and the constructed constrained variational problem is presented as follows:

$$\min_{\{u_k\}, \{\omega_k\}} \left\{ \sum_{k=1}^K \|\partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t}\|_2^2 \right\} \quad (12)$$

$$s.t. RV(t) = \sum_{k=1}^K u_k(t)$$

where $\{u_k\}$ and $\{\omega_k\}$ represent the congregation of k modal components and their corresponding central frequencies after decomposition, respectively. $\partial(\bullet)$ denotes the partial derivative of time t , $\delta(t)$ is the impulse function, and $*$ indicates the convolution symbol.

(ii) First of all, the quadratic penalty factor α and Lagrange multiplier $\lambda(t)$ are introduced, and the above Eq. (12) is transformed into a non-constrained variational problem. The transformed augmented Lagrangian function expression is:

$$L(\{u_k(t)\}, \{\omega_k\}, \lambda) = \alpha \sum_{k=1}^K \|\partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t}\|_2^2 + \|RV(t) - \sum_{k=1}^K u_k(t)\|_2^2 + \left\langle \lambda(t), RV(t) - \sum_{k=1}^K u_k(t) \right\rangle \quad (13)$$

Secondly, the alternate direction method of multipliers (ADMM) is applied to alternately update $\hat{u}_k^{n+1}(\omega)$, $\hat{\omega}_k^{n+1}$ and $\hat{\lambda}_k^{n+1}(\omega)$ (Hestenes, 1969). The updating process is as follows:

$$\hat{u}_k^{n+1}(\omega) = \frac{\widehat{RV}(\omega) - \sum_{i \neq k} \hat{u}_i(\omega) + \hat{\lambda}(\omega)/2}{1 + 2\alpha(\omega - \omega_k)^2} \quad (14)$$

$$\hat{\omega}_k^{n+1} = \frac{\int_0^\infty \omega |\hat{u}_k^{n+1}(\omega)|^2 d\omega}{\int_0^\infty |\hat{u}_k^{n+1}(\omega)|^2 d\omega} \quad (15)$$

$$\hat{\lambda}_k^{n+1}(\omega) = \hat{\lambda}_k^n(\omega) + \zeta \left[\widehat{RV}(\omega) - \sum_{k=1}^K \hat{u}_k^{n+1}(\omega) \right] \quad (16)$$

where n represents the number of iterations, $\widehat{RV}(\omega)$, $\lambda^n(\omega)$, $\hat{u}_i(\omega)$ and $\hat{u}_k^{n+1}(\omega)$ are the Fourier transform of $RV(t)$, $\lambda^n(t)$, $u_i(t)$ and $u_k^{n+1}(t)$, respectively, ζ is the noise tolerance. The updating process is repeated until the convergence accuracy ε satisfies the $\sum_{k=1}^K \|\hat{u}_k^{n+1} - \hat{u}_k^n\|_2^2 / \|\hat{u}_k^n\|_2^2 < \varepsilon$.

Finally, $u_k(t)$ ($k = 1, 2, \dots, K$) are obtained.

2.1.3. RV sequence decomposition method based on GWO-VMD

Considering the irregularity and complexity of the RV sequence, VMD can decompose the series into multiple IMFs with different economic meanings to facilitate the neural network to capture the features contained in the series more effectively. Therefore, it is applicable for the decomposition of stock price index volatility.

During the process of decomposing the RV series of stock price index by VMD, the total number of decomposed modes k and the quadratic penalty coefficient α will directly affect the decomposition results of the $RV(t)$. Specifically, on the one hand, appropriate k can fully separate the modes of $RV(t)$ and prevent the occurrence of mode mixing. On the other hand, appropriate α can improve the reconstruction efficiency of $RV(t)$. Accordingly, the parameters k and α of VMD are optimized by GWO algorithm.

It is necessary to remark that since the envelope entropy has the capability to measure the periodicity and sparsity of IMF, the smaller the envelope entropy value is when the IMF has less noise and richer features (Niu, Sun, Yu, & Wang, 2022). Accordingly, the minimum envelope entropy is selected as the fitness function of GWO algorithm to guide the parameter selection of VMD. The central consideration of envelope entropy is to obtain the envelope signal after Hilbert transform and extract the signal characteristics by combining the information entropy. When the signal is demodulated by Hilbert, the original signal sequence is transformed into a probability distribution sequence. The envelope entropy $E_k(t)$ is mathematically defined as:

$$E_k(t) = - \sum_{k=1}^K p_k(t) \lg p_k(t) \quad (17)$$

$$p_k(t) = g_k(t) / \sum_{k=1}^K g_k(t) \quad (18)$$

where $g_k(t)$ is the envelope signal of $u_k(t)$ after Hilbert demodulation, $p_k(t)$ is the normalized form of $g_k(t)$.

The GWO algorithm is employed to search for the best combination of parameters for the VMD algorithm, and the steps are as follows:

Step1 Initialize the parameters of GWO.

Step2 Decompose the RV series of stock price index through VMD, and calculate the envelope entropy value of (α) wolf, (β) wolf and (δ) wolf.

Step3 Update the position of the current wolves using Eqs. (7) to (9).

Step4 Adjust the position of the wolves outside the search space, and recalculate the envelope entropy value for all wolves.

Step5 If iteration number l is less than the maximum number of iterations, repeat **Steps 2 to 5** in order to compare and continuously update the minimum envelope entropy value corresponding to each solution. Otherwise, stop the iterations.

Step6 Output the position of (α) wolf, which is to obtain the optimal parameter combination $[k, \alpha]$.

2.2. Forecasting method based on deep reinforcement learning

2.2.1. Deep belief network

On the ground of deep learning theory, Hinton and Salakhutdinov (2006) proposed a DBN model to strengthen the ability of neural networks to solve complex nonlinear problems. The DBN is a probabilistic generative model which consists of multi-layer stacked restricted Boltzmann machine (RBM) and logistic regression layers. As can be observed from the topological structure of RBM displayed in Fig. 2, it includes a visible layer v and a hidden layer h . In detail, there is a bidirectional fully connected between the neurons in the visible layer and the hidden layer, the former serves as the input of RV sequence, the latter serves with extracting the characteristics of RV. v_i and h_j are each denoted as i -th and j -th neurons of h and v . w^p is expressed as the connection weight matrix between the two layers, b^h and b^v stand for the bias of h and v , respectively.

As shown in Fig. 3, the DBN forecasting model contains visible layer, hidden layer and output layer. First of all, the input data are received by the visible layer of RBM₁ and transmitted to the hidden layer, which learns layer by layer adopting a greedy algorithm with unsupervised learning characteristics. In this way, the inner relationship of the sample data are continuously output to the hidden layer of the next RBM. Eventually, the output of RBM_m are received through the supervised learning method of BP, and the whole network weights are fine-tuned by the back propagation algorithm.

Therefore, the training of DBN model includes two parts: unsupervised pre-training and supervised fine-tuning.

(i) Pre-training process.

For the given visible layer and hidden layer, the energy function of RBM can be calculated:

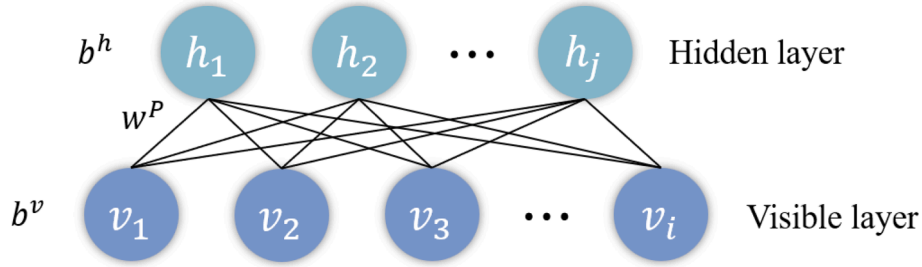


Fig. 2. Structure diagram of RBM.

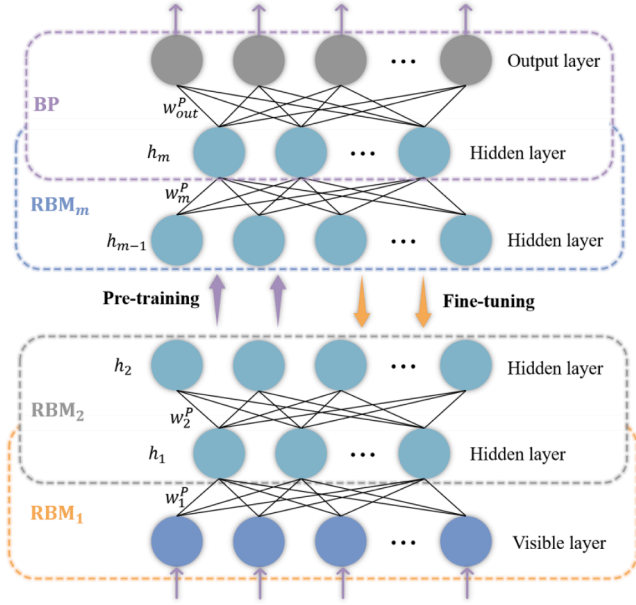


Fig. 3. Architecture of DBN model.

$$E(v, h|\theta) = -\sum_{i=1}^n \sum_{j=1}^m w_{ij}^p v_i h_j - \sum_{i=1}^n b_i^v v_i - \sum_{j=1}^m b_j^h h_j \quad (19)$$

where n and m separately indicate the number of neurons in v and h , $\theta = \{w^p, b^v, b^h\}$ denotes all parameters of RBM.

The joint probability distribution function of v and h is determined by Boltzmann distribution:

$$P(v, h|\theta) = \exp(-E(v, h|\theta)) / \sum_v \sum_h \exp(-E(v, h|\theta)) \quad (20)$$

In the RBM model, the visible nodes and hidden nodes are independent when $h_j = 1$ or $v_i = 1$. Consequently, the probability of visible layer and hidden layer are defined as Eq. (21) and Eq. (22), respectively.

$$P(h_j = 1|v, \theta) = \text{sigm}\left(\sum_i w_{ij}^p v_i + b_j^h\right) \quad (21)$$

$$P(v_i = 1|h, \theta) = \text{sigm}\left(\sum_j w_{ij}^p h_j + b_i^v\right) \quad (22)$$

where $\text{sigm}(\bullet)$ denotes the sigmoid function.

The set of parameters θ is updated by using the Contrastive Divergence (CD) algorithm with the following rules:

$$\Delta w_{ij}^p = \eta(\langle v_i h_j \rangle_{data} - \langle v_i h_j \rangle_{rec}) \quad (23)$$

$$\Delta b_i^v = \eta(\langle v_i \rangle_{data} - \langle v_i \rangle_{rec}) \quad (24)$$

$$\Delta b_j^h = \eta(\langle h_j \rangle_{data} - \langle h_j \rangle_{rec}) \quad (25)$$

where η is the learning rate, $\langle \bullet \rangle_{data}$ is the expectation of the input sample data, $\langle \bullet \rangle_{rec}$ denotes the expectation of the reconstructed data distribution learned by the model.

(ii) Fine-tuning process.

In order to overcome the problem that parameters are easy to fall into local optimum, back propagation algorithm is adopted to optimize the weight $W^p = (w_1^p, w_2^p, \dots, w_m^p, w_{out}^p)$ obtained by pre-training, and then the parameters of the whole DBN are adjusted from top to bottom and from bottom to top. Finally, the final weight $W = (w_1, w_2, \dots, w_m, w_{out})$ of DBN is obtained. Supposing $o(t)$ and $\hat{o}(t)$ are the real output and expected output of DBN, respectively. The loss function of output layer is:

$$L(\vartheta) = \sum_{t=1}^T (\hat{o}(t) - o(t))^2 / 2 \quad (26)$$

where ϑ stands for the number of iterations, T represents the length of $o(t)$.

The weight update function between the output layer and last hidden layer can be indicated as follows:

$$\Delta w_{out} = w_{out}(\vartheta + 1) - w_{out}(\vartheta) = -\eta' \partial L(\vartheta) / \partial w_{out}(\vartheta) \quad (27)$$

where η' is the learning rate of BP, $w_{out}(\vartheta + 1)$ is the connection weight after reverse fine-tuning. $\partial L(\vartheta) / \partial w_{out}(\vartheta)$ is the gradient of the w_{out} parameter, which is calculated by its error backward diffusion.

2.2.2. Long short-term memory

As a recurrent neural network architecture, LSTM not only avoids long-term dependence, but also addresses the problem of gradient approaching zero (vanishing) or infinite (exploding). As Fig. 4 illustrates, LSTM is mainly consisted of three gate control mechanisms with

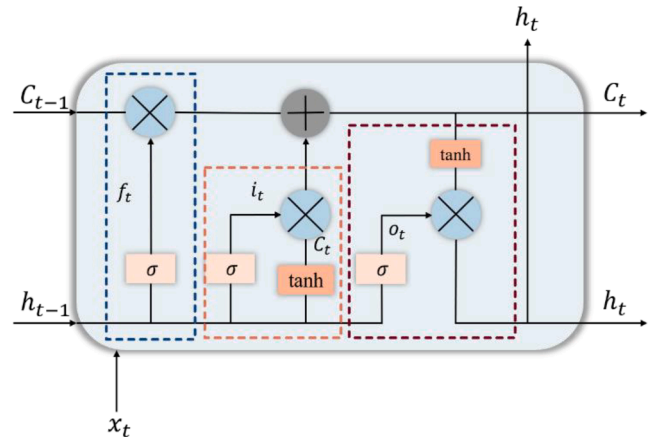


Fig. 4. The structure of cell state in LSTM network.

different input types, namely input gate i_t , forget gate f_t and output gate o_t . The input gate is responsible for determining the information stored in the current memory cell state, and the output gate affects the output of the current state h_t . The forget gate makes use of the information from the last memory cell C_{t-1} that was selectively forgotten.

The cell state and hidden state are updated through a set of mathematical calculations (see Eq. (28) to Eq. (33)), enabling the network to track long-term patterns and provide reliable forecasting according to the historical data.

$$f_t = \text{sigm}(W_f[h_{t-1}, x_t] + b_f) \quad (28)$$

$$i_t = \text{sigm}(W_i[h_{t-1}, x_t] + b_i) \quad (29)$$

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (30)$$

$$C_t = f_t \circ C_{t-1} + i_t \circ \tilde{C}_t \quad (31)$$

$$o_t = \text{sigm}(W_o[h_{t-1}, x_t] + b_o) \quad (32)$$

$$h_t = o_t \circ \tanh(C_t) \quad (33)$$

where W_f, W_i, W_o are the weight matrix of f_t, i_t, o_t , respectively. b_f, b_i, b_o denote the bias of f_t, i_t, o_t , respectively. $\tilde{C}_t$ is the new candidate state. h_{t-1} is the output value of the LSTM unit at the previous moment. x_t is the input of the current time t . C_t is the current memory cell. $\circ$ is the Hadamard product, $\text{sigm}(x) = 1/(1 + e^{-x})$ denotes the sigmoid activation function, and $\tanh(x) = (e^x - e^{-x})/(e^x + e^{-x})$ is the nonlinear mapping function.

In the initial stage of LSTM operation, the forget gate selects to keep or forget the content of the previous cell state in accordance with the sigmoid activation function for preserving long-distance information. Next, in order to update the internal cell state of LSTM, the product of the forget gate output and the previous cell state is added to the product of the input gate output and the candidate cell state. Subsequently, the output gate controls the output to the next cell in the LSTM structure by using the sigmoid activation function. At the end, the output of the output gate is multiplied with the regulated output of the hyperbolic tangent function that updates the cell state, and the final value is incorporated into hidden state. The cell state and output will be forwarded to the next step, and the LSTM operation will continue to repeat the same process as above.

2.2.3. Gated recurrent unit

GRU is a novel RNN gating mechanism which can memorize the historical information. As shown in Fig. 5, GRU has two gating units, the reset gate r_t and the update gate z_t (the combination of input gate and forget gate), which significantly reduces the number of parameters and thereby dramatically enhances the efficiency of the model. The reset

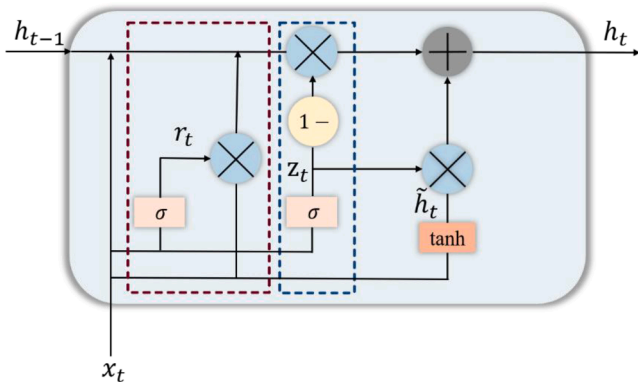


Fig. 5. The structure of cell state in GRU network.

gate r_t controls the degree of merging the current state information with the previous state information, and the update gate z_t controls whether the state information at the previous moment is retained and the degree of retention in the current state. If the output at the previous time is h_{t-1} and the input at the current time is x_t , the process of computing the hidden layer unit h_t based on GRU can be calculated by the following formulas Eq. (34) to Eq. (37):

$$r_t = \text{sigm}(W_r[h_{t-1}, x_t] + b_r) \quad (34)$$

$$z_t = \text{sigm}(W_z[h_{t-1}, x_t] + b_z) \quad (35)$$

$$\tilde{h}_t = \tanh(W_h[r_t \circ h_{t-1}, x_t] + b_h) \quad (36)$$

$$h_t = (1 - z_t) \circ h_{t-1} + z_t \circ \tilde{h}_t \quad (37)$$

where W_r, W_z, W_h are the weight matrix of $r_t, z_t, \tilde{h}_t$, respectively. b_r, b_z, b_h denote the bias of $r_t, z_t, \tilde{h}_t$, respectively.

At first, the hidden layer state that cannot be retained at the current moment is controlled by the reset gate. Secondly, the hidden layer state that remains at the current time is determined by the update gate. Finally, the hidden layer candidate state is calculated according to the hidden layer state of the previous moment and the input of the current moment, and the forward hidden layer state of the current moment output is obtained by combining the output of the previous moment. GRU is trained to be capable of some forecasting works, which can reduce the risk of overfitting due to the small number of parameters.

2.2.4. Q-learning algorithm

The Q-learning algorithm constructs the target state matrix S and action state matrix A into a table to store the Q-value, and afterwards selects the action that can obtain the maximum benefit in accordance with the Q-value. Consequently, our study adopts the Q-learning algorithm as an integrated learning method, that is, the Q-learning algorithm assigns weights to three deep networks on the basis of setting reasonable rewards and punishment strategies, and the concrete implementation process is as follows:

Step1 Create matrix $S = [\omega_i^d, \omega_i^l, \omega_i^g]$ and matrix $A = [\Delta\omega_i^d, \Delta\omega_i^l, \Delta\omega_i^g]$. Where the elements in matrix S separately denote the weight coefficients of DBN, LSTM and GRU. The elements in matrix A separately represent the magnitude of the adjustment action.

Step2 Construct the loss function E and reward mechanism R :

$$R = \begin{cases} E_t - E_{t+1} - 1, & E_t < E_{t+1} \\ E_t - E_{t+1} + 1, & E_t \geq E_{t+1} \end{cases} \quad (38)$$

$$E = \sqrt{T^{-1} \sum_{t=1}^T [u_i(t) - y_i(t)]^2} \quad (39)$$

where T denotes the number of input samples, $y_i(t)$ represent the predicted values of i -th IMF.

Step3 Based on ϵ -greedy mechanism, the agent selects the optimal action with the probability of $1 - \epsilon$, and randomly selects the action with the probability of ϵ . The action is selected as follows:

$$a_m = \begin{cases} \text{on account of } \max Q, & P = 1 - \epsilon \\ \text{random } a \in A, & P = \epsilon \end{cases} \quad (40)$$

where $\epsilon \in (0, 1)$ is the exploration probability.

Step4 Calculate E and R , and update the Q-value table using the calculated value function $Q_t(S_t, a_t)$ with the following procedure:

$$Q_{t+1}(S_t, a_t) = Q_t(S_t, a_t) + \mu \left(r_t + \phi \max_a Q_t(S_{t+1}, a_{t+1}) - Q_t(S_t, a_t) \right) \quad (41)$$

where μ is the learning rate, ϕ is the discount factor, repeat **steps 3 to 4** until the iteration is terminated.

2.3. Framework of the proposed RV sequence forecasting model

In this subsection, the overall framework of the proposed model is shown in Fig. 6, which includes four parts: adaptive decomposition part, deep neural network prediction part, Q-learning optimization part, summary and analysis part. The specific modeling process of our proposed model is described below:

Part1 Adaptive decomposition

According to the VMD parameter combination $[k, \alpha]$ optimized by the GWO algorithm, $RV(t)$ is decomposed into $u_k(t)$ ($k = 1, 2, \dots, K$), and then normalized and divided into mutually exclusive sets, that is, the training set and the testing set. The training set is used to train the parameters of the model, and the testing set is used to verify the generalization performance of the trained final model.

Part2 Deep neural network prediction

Taking the i -th subsequence $u_i(t) = \{x_1, x_2, \dots, x_T\}$ decomposed by

$$\tilde{p}_i(T+h) = \begin{cases} \tilde{M}[p_i(T), \dots, p_i(T-w+1)], & \text{if } h = 1 \\ \tilde{M}[\tilde{p}_i(T+h-1), \dots, \tilde{p}_i(T+1), p_i(T), p_i(T-w+1)], & \text{if } h \in \{2, \dots, w\} \\ \tilde{M}[\tilde{p}_i(T+h-1), \dots, \tilde{p}_i(T+h-w)], & \text{if } h \in \{w+1, \dots, H\} \end{cases} \quad (42)$$

VMD technique as an example, we consider the mean squared error (MSE) as the loss function and construct a forecasting model by utilizing three deep neural networks DBN, LSTM and GRU to output the predicted sequences $p_i^d(t)$, $p_i^l(t)$ and $p_i^g(t)$.

It should be noted that this paper adopts a sliding time window for rolling prediction, which can better fit the time-varying behavior and capture the time changes of related features (Zhang, & Zhang, 2021). Assuming that the current window length of the sliding window is w , the single-step prediction model can be expressed as $p_i(j+1) =$

$M[p_i(j), \dots, p_i(j-w+1)] + e$, $j \in \{w, \dots, T-1\}$, where e represents the random term that contains interference and noise. As a result, the training input matrix is $[P_{i,1}, P_{i,2}, \dots, P_{i,w}]$, where $P_{i,s} = [p_i(j), \dots, p_i(T-w-1+s)]^T$, and the training output matrix is $P_{i,w+1} = [p_i(w+1), \dots, p_i(T)]^T$. In order to maintain the integrity of the article, we next explain the principle for the multi-step prediction of the robustness check in Section 3.6.

In this paper, we use a recursive strategy for multi-step prediction, which is simple, intuitive, and fast to run by training only a single-step prediction model (Herrera, Pomares, Rojas, Guillén, Prieto, & Valenzuela, 2007; Yuan, & Che, 2022). The $p_i(j+1)$ obtained from the prediction is used as next "new input", which is then substituted into Eq. (42) for the next prediction. This process is iterated until all prediction ranges of the RV series is estimated. Recursive multi-step prediction can be defined as follows.

where $\tilde{M}$ represents the single model. $\tilde{p}_i(T+h)$ denotes the h -step prediction of i -th IMF. H stands for the size of prediction step.

If $h = 1$, all the input variables are original values, and the prediction accuracy is optimal; If $h \in \{2, \dots, w\}$, the previous predicted values need to be substituted into the prediction model as input, while the initial original values are discarded to ensure the consistency of the embedded dimension, so as to carry out multi-step prediction; If $h \in \{w+1, \dots, H\}$, all input variables are estimated values.

Part3 Q-learning optimization

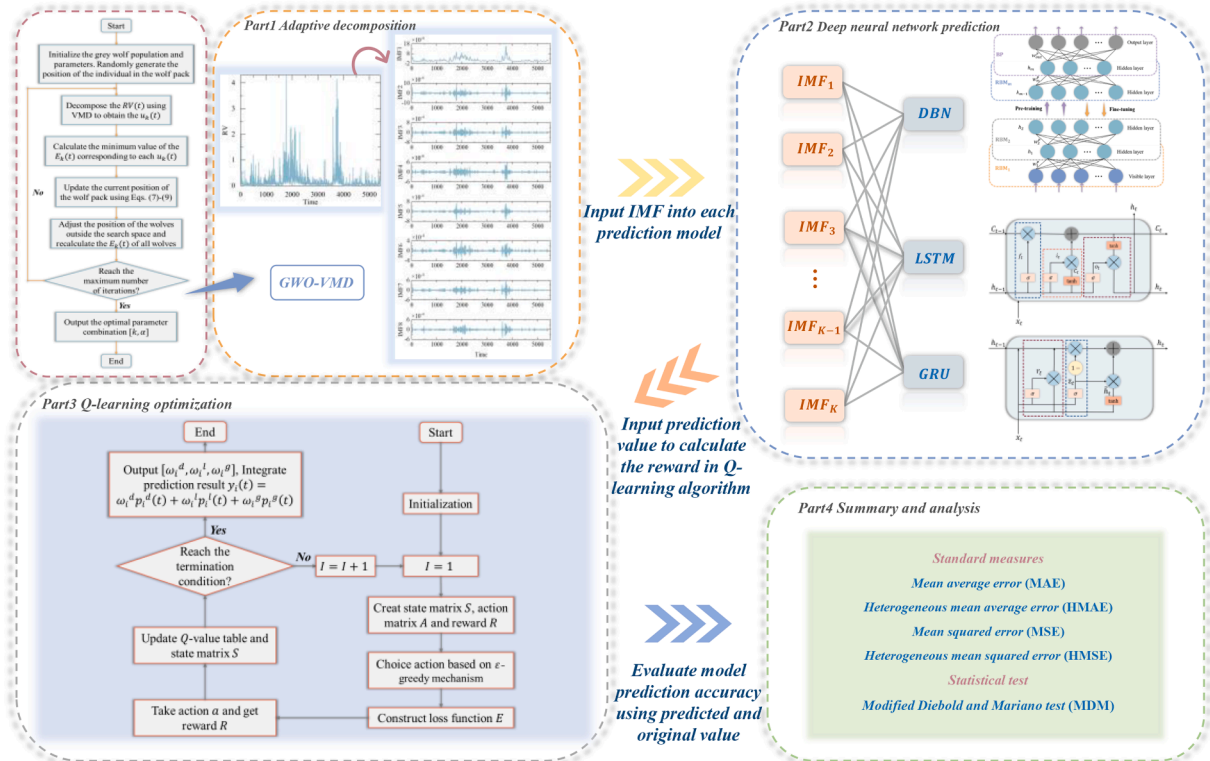


Fig. 6. The overall framework of the proposed GVMD-Q-DBN-LSTM-GRU model.

The Q-learning algorithm obtains the optimal weight combination $[\omega_i^d, \omega_i^l, \omega_i^g]$ of $p_i^d(t)$, $p_i^l(t)$ and $p_i^g(t)$ by analyzing the prediction ability of each deep neural network, and acquires the predicted result of $u_i(t)$ through $y_i(t) = \omega_i^d p_i^d(t) + \omega_i^l p_i^l(t) + \omega_i^g p_i^g(t)$.

Part4 Summary and analysis

The ultimate result $\widehat{RV}(t) = \sum_{k=1}^K y_k(t)$ is obtained by using addition to integrate the forecasting results of $u_k(t)$. In this study, not only four evaluation indexes, MAE, MSE, HMAE and HMSE, are selected to evaluate the difference between the real and predicted values of $RV(t)$, but also the MDM statistical test is adopted in order to measure whether there is a significant difference in the prediction effect between the GVMD-Q-DBN-LSTM-GRU model and the comparison model.

3. Empirical analysis

This section first introduces the source of empirical data and their specific characteristics, and further elaborates the evaluation indicators. Secondly, the settings and results of VMD method, baseline models and Q-learning algorithm parameters are explained. Finally, we perform comparative experiments on different horizons to demonstrate the superiority and robustness of the proposed GVMD-Q-DBN-LSTM-GRU model. It should be specified that the hyperparameters of the baseline model are the optimal values obtained from the experiments.

3.1. Dataset

In our paper, three stock price indices from diverse regions are selected to verify the forecasting power of the proposed GVMD-Q-DBN-LSTM-GRU model on different datasets. The first dataset is the SSEC index for China, which includes a capital stock weighted index of all listed companies in Shanghai and has emerged as a commonly adopted indicator in Chinese stock market, thus being categorized as an emerging market. Differently, the second dataset is the SPX index from the US, which is considered to be the developed market in the world owing to the mature and complete market operation system of the US stock market. The third dataset is the FTSE index for the UK, which is the most authoritative and representative important stock price index in the UK. As another developed market, the UK stock market is the largest stock market in Europe and the most international one in the world.

In fact, the maturity of the financial market may affect the validity of the prediction model (Deng, Huang, Hasan, & Bao, 2022). Therefore, this paper selects three stock price indices: SSEC, SPX and FTSE, covering the sample period from January 2000 to June 2022. Moreover, since 5-min high-frequency data provide a balance between measurement accuracy and mitigation of market microstructure noise (Andersen, & Bollerslev, 1998), we take 5 min as the sampling interval to calculate the daily RV of three stock price indices. Specifically, the SSEC sample contains 5427 trading days, from January 4, 2000 to June 28, 2022; the SPX sample contains 5635 trading days, from January 3, 2000 to June 28, 2022; the FTSE sample contains 5668 trading days, from January 4, 2000 to June 28, 2022. To facilitate obtaining a first impression of the dataset, we first draw the RV sequences of three stock price indices (see Fig. 7) and their descriptive statistics are presented in Table 1. In Fig. 7, the horizontal coordinate represents the number of sample points, and the vertical coordinate represents the RV of different stock price indices.

From Table 1, it can be observed that the above datasets all exhibit high time-varying characteristics, in other words, the standard deviation of the dataset is very large compared with the sample average, and all show excessive kurtosis. These results indicate that RV sequence does not follow a normal distribution. In addition, the K-S statistic rejects the null hypothesis of normal distribution at the 1% significance level, which also confirms the above observation results. The Ljung-Box test shows that at the 1% significance level, the null hypothesis is rejected, indicating that the RV sequence all have 10th-order lagged

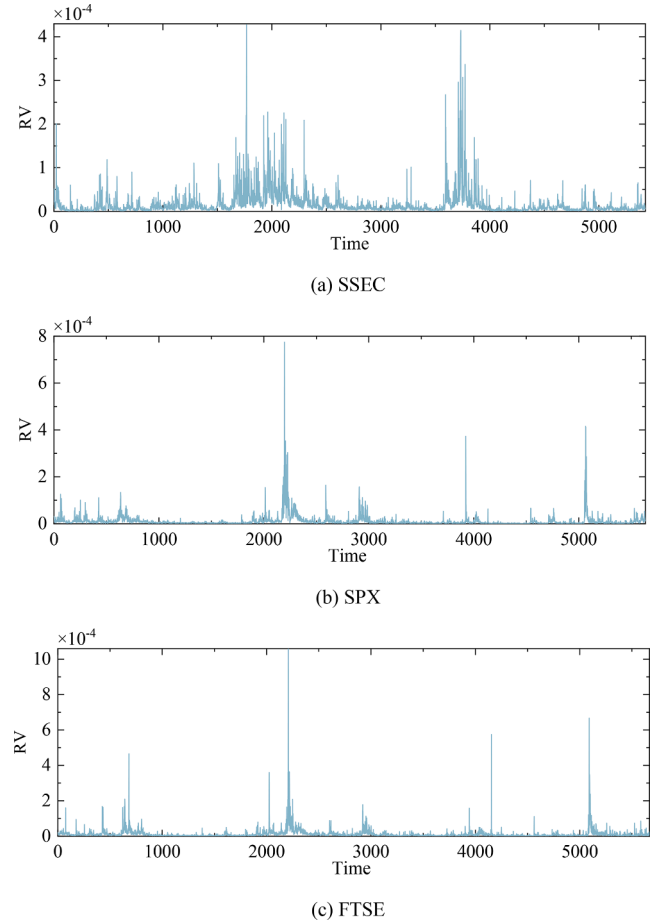


Fig. 7. Original RV time sequence.

Table 1

Descriptive statistics for the target RV sequence.

Index	SSEC	SPX	FTSE
Mean	1.5462E-04	1.0957E-04	1.2012E-04
Std	2.6049E-04	2.5746E-04	2.8815E-04
Minimum	3.5433E-06	1.2181E-06	1.3324E-06
Maximum	4.2786E-03	7.7477E-03	1.0600E-02
Kurtosis	62.2050	203.4472	420.2390
Skewness	6.2635	11.0025	15.9394
K-S	0.2830***	0.3370***	0.3440***
Q(10)	1.3217E + 04***	1.7997E + 04***	9.1238E + 03***

Note: *, ** and *** denote rejection of the null hypothesis at the 10%, 5% and 1% significance level respectively. Std denotes the standard deviation. K-S represents the Kolmogorov-Smirnov statistics. Q(10) is the Ljung-Box statistic of the RV for up to the 10th order serial correlation.

autocorrelation and long memory.

3.2. Forecasting evaluation criteria

With the purpose of evaluating and comparing the performance of various RV sequence forecasting models, the MAE, MSE and their relative indexes, namely HMAE and HMSE, are applied as evaluation metrics in this paper with the following equations (Fuentes, Izzeldin, & Kalotychou, 2009).

$$MAE = T^{-1} \sum_{t=1}^T |RV(t) - \widehat{RV}(t)| \quad (43)$$

$$MSE = T^{-1} \sum_{t=1}^T (RV(t) - \widehat{RV}(t))^2 \quad (44)$$

$$HMAE = T^{-1} \sum_{t=1}^T |1 - (\widehat{RV}(t)/RV(t))| \quad (45)$$

$$HMSE = T^{-1} \sum_{t=1}^T [1 - (\widehat{RV}(t)/RV(t))]^2 \quad (46)$$

where T represents the number of samples in the testing set, $RV(t)$ and $\widehat{RV}(t)$ are the real value and predicted value of the testing set of RV sequence (the first 80 % of the total sample is selected as the training set, and the remaining 20 % is the testing set). In addition, this study uses the percentage of promotion for MAE (P_{MAE}), percentage of promotion for MSE (P_{MSE}), percentage of promotion for HMAE (P_{HMAE}), and percentage of promotion for HMSE (P_{HMSE}) for direct comparison of the predictive capability between two models, which defined in Eqs. (47)-(50).

$$P_{MAE} = (MAE_1 - MAE_2)/MAE_1 * 100\% \quad (47)$$

$$P_{MSE} = (MSE_1 - MSE_2)/MSE_1 * 100\% \quad (48)$$

$$P_{HMAE} = (HMAE_1 - HMAE_2)/HMAE_1 * 100\% \quad (49)$$

$$P_{HMSE} = (HMSE_1 - HMSE_2)/HMSE_1 * 100\% \quad (50)$$

As a matter of fact, the loss function is susceptible to the interference of outliers in the sequence, which leads to a certain degree of limitation in the judgment of the model prediction performance (Hansen, & Lunde, 2005). Therefore, this paper further makes use of the MDM statistical

test to measure whether the prediction effect between the proposed model and the comparison model is significant. Assuming that the predicted values of the two comparison models are expressed as $\{\widehat{RV}_l(t), l = 1, 2, t = 1, 2, \dots, T\}$, where T is the length of sequence. Then the prediction error $\{e_l(t) = RV(t) - \widehat{RV}_l(t), l = 1, 2\}$ can be determined by the loss function $L(\bullet)$ at time t , and the loss function related to $\{\widehat{RV}_l(t)\}$ can be expressed as $\{L(e_l(t)), l = 1, 2\}$. Loss difference is defined as follows:

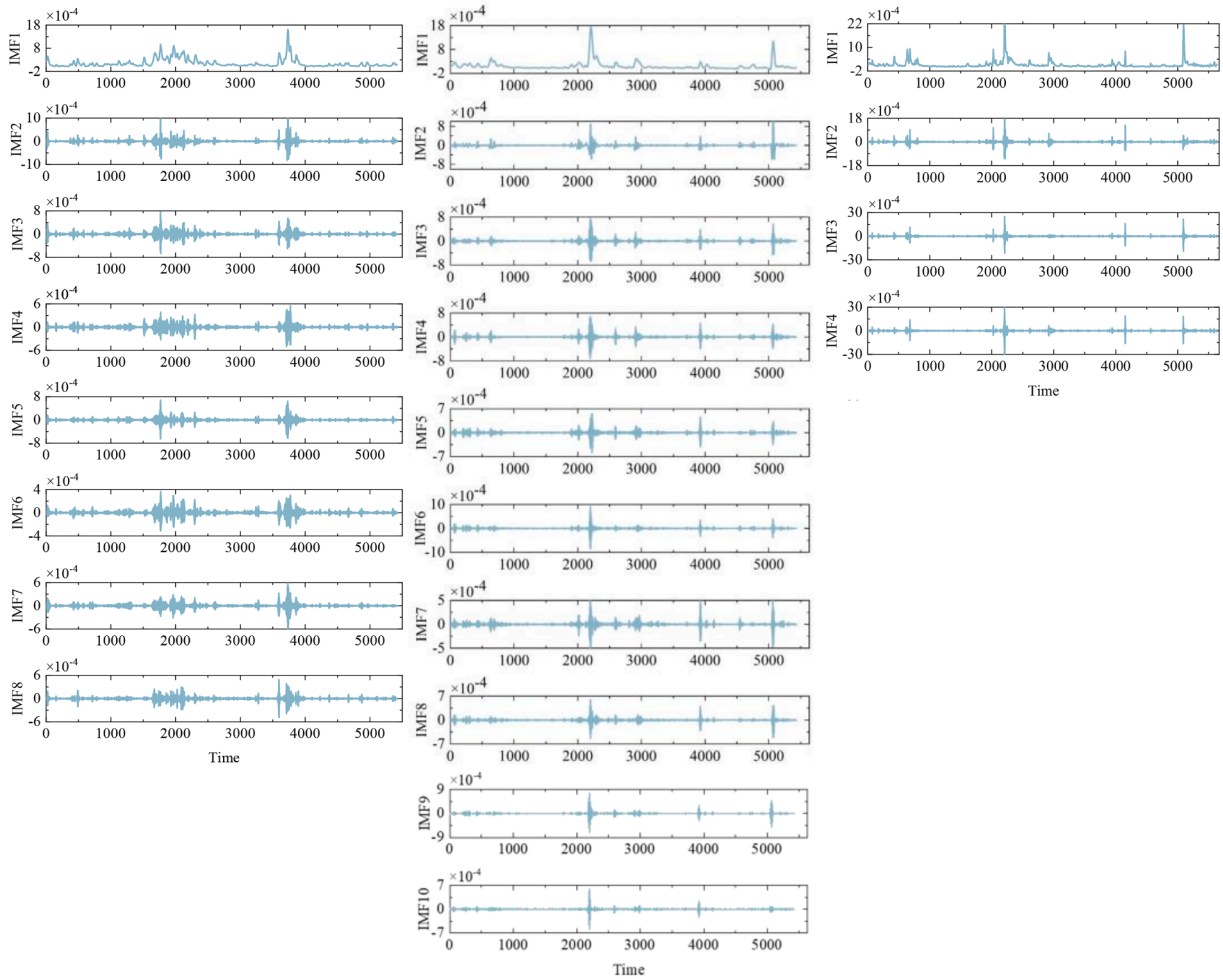
$$d_t = L(e_1(t)) - L(e_2(t)), t = 1, 2, \dots, T \quad (51)$$

The null hypothesis of the MDM statistical test is $H_0 : E(d(t)) = 0, t = 1, 2, \dots, T$, which indicates that two models have the same prediction accuracy, and the alternative hypothesis is $E(d(t)) \neq 0$, which indicates that there is a difference in the prediction performance between the models. Under the assumption of stationary $d(t)$ covariance and short-term memory, DM presents the test of H_0 in the Eq. (52) below:

$$\sqrt{T}(\bar{d} - E(d(t))) \xrightarrow{d} N(0, \omega^2) \quad (52)$$

where $\bar{d} = T^{-1} \sum_{t=1}^T d(t)$, ω^2 represents the long memory variance of $d(t)$, its expression is $\omega^2 = \sum_{j=-\infty}^{\infty} \gamma(j), \gamma(j) = Cov(d(t), d(t-j))$, $\hat{\omega}^2$ improves the continuous long-term variance estimator ω^2 , which expressed as:

$$\hat{\omega}^2 = \hat{\gamma}(0) + 2 \sum_{j=1}^{h-1} \hat{\gamma}(j) \quad (53)$$



(a) SSEC

(b) SPX

(c) FTSE

Fig. 8. Results of GWO-VMD based RV sequence decomposition.

where $\hat{\gamma}(j) = T^{-1} \sum_{t=j+1}^T (d(t) - \bar{d})(d(t-j) - \bar{d})$. The purpose of the MDM test is to improve the test behavior of a small sample size, and to use the student t critical value instead of the standard normal distribution. The MDM statistic is expressed as follows:

$$\text{MDM} = \sqrt{T+1-2h+T^{-1}h(h-1)}(\bar{d}/\hat{\omega}) \quad (54)$$

The optimal h - step prediction error depends at most on $h-1$ and lags behind $h-1$.

3.3. Determination of VMD optimal parameter combination

As presented in Fig. 7, the RV series of the stock price index fluctuate greatly, which indicates a large mutual interference between signals of dissimilar frequencies in the RV. With the purpose of promoting the stability and accuracy of forecasting model, GWO-VMD is adopted in this paper to decompose the RV sequences with intense fluctuations into several sub-series characterized by distinct frequencies.

More further, to deal with the impact of parameter determination on the decomposition effect of VMD, the maximum number of iterations of GWO was set to 30, the number of search agents was set to 20, and the upper and lower limits of parameter k and α values were set to [3, 500] and [15, 2500], respectively. Fig. 8 displays the superlative decomposition results for three diverse datasets SSEC, SPX and FTSE, and their corresponding optimal values of k and α are [8, 2019], [10, 2242] and [4, 726], separately. Without a doubt, it is noticeable that the IMFs of dissimilar frequencies after decomposition are more uniform in time domain, indicating that VMD has the ability to fully extract the characteristics of RV series.

3.4. Determination of optimal hyperparameters

3.4.1. Settings of baseline model

In this paper, ten diverse prediction models are selected to compare with the proposed GVMD-Q-DBN-LSTM-GRU model, including BPNN, extreme learning machine (ELM), support vector regression (SVR), DBN, LSTM, GRU, DBN-LSTM-GRU, GVMD-DBN-LSTM-GRU, Q-DBN-LSTM-GRU and WT-Q-DBN-LSTM-GRU. On this basis, the forecasting outcomes of variant framework models are compared. In a single model, DBN, LSTM and GRU show the excellent prediction performance, thus we select the above three models as the baseline model and experiment with different combinations of input parameters to obtain the optimum selection of base model parameters. The Hyperparameters are given in Table 2.

3.4.2. Settings of Q-learning algorithm

In order to combine the three deep neural networks (DBN, LSTM and GRU) into an integrated model with excellent prediction performance and obtain the best generalization performance, our paper applies the Q-learning algorithm to allocate the weight to three deep neural networks. However, the use of Q-learning algorithm inevitably requires decisions on the number of episode during network finalization. Therefore, our study adopts the Q-learning algorithm with the Early Stopping method

during the training process, which is implemented to address the problem that the number of episode needs to be set manually (Prechelt, 1998). The stopping method of the episode is set to autonomous stop, which limits the number of episode required for loss minimization (Wan, Wu, Cao, & Wu, 2023; Egrioglu, & Bas, 2023).

During the iteration process of the Q-learning algorithm, the number of episode ('episode') is set to 10,000 and the tolerance period ('max-period') is set to 500. For the integrated models Q-DBN-LSTM-GRU, WT-Q-DBN-LSTM-GRU and GVMD-Q-DBN-LSTM-GRU mode after integration by the Q-learning algorithm are trained at each time with the value

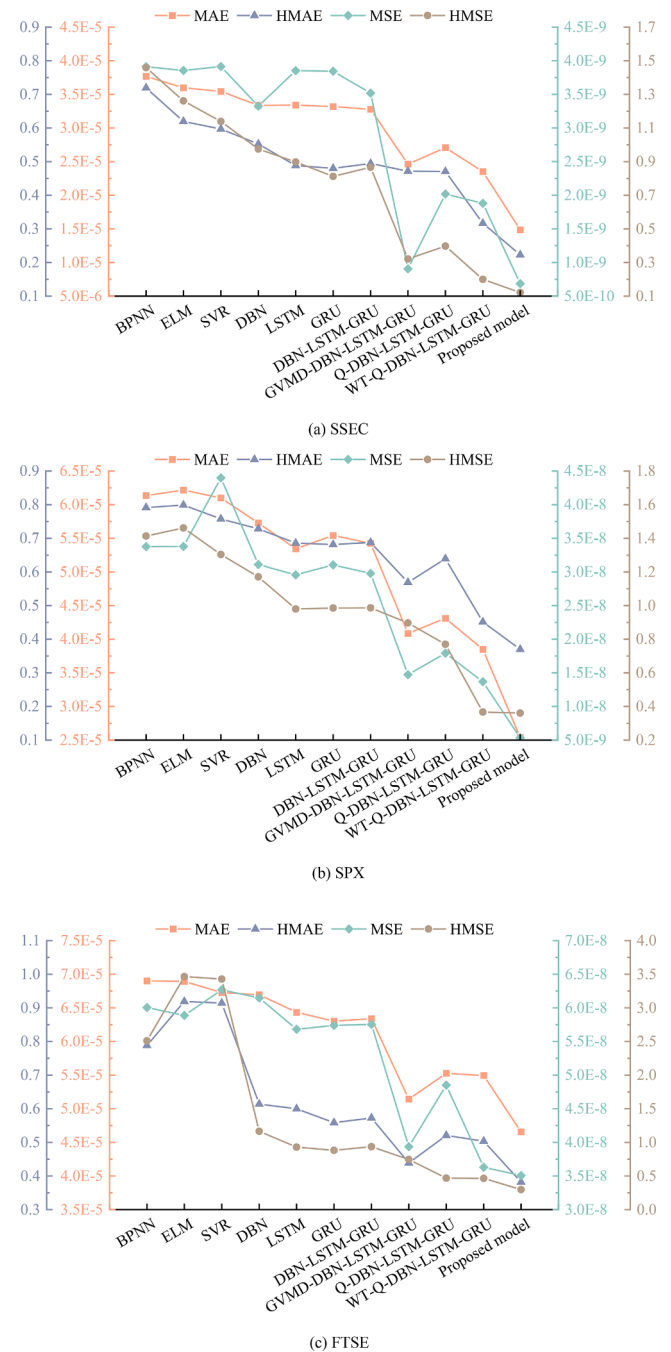


Fig. 9. The evaluation values of each model on three datasets.

Table 2
Hyperparameter settings selection of baseline models.

Model	Name of Parameter	SSEC	SPX	FTSE
DBN	Number of the Hidden units	[25,10]	[50,20]	[25,10]
	Learning rate in Pre-training process	0.01	0.0001	0.0001
	Learning rate in Fine-tuning process	0.0001	0.0001	0.0001
LSTM	Number of the Hidden units	256	32	64
	Optimizer	Adam	Adam	Adam
	Dropout	0.2	0.2	0.3
GRU	Number of the Hidden units	128	32	32
	Optimizer	Adam	Adam	Adam
	Dropout	0.2	0.2	0.2

function Q satisfying convergence within the specified number of episodes, including the maximum number of 9662 episodes and the minimum number of 116 episodes.

Table 3

The deterministic forecasting results of the GVMD-Q-DBN-LSTM-GRU model and compared models.

Dataset	Model	MAE	MSE	HMAE	HMSE
SSEC	BPNN	3.7682E-05	3.9109E-09	0.7193	1.4588
	ELM	3.5980E-05	3.8552E-09	0.6194	1.2612
	SVR	3.5441E-05	3.9141E-09	0.5969	1.1390
	DBN	3.3364E-05	3.3250E-09	0.5523	0.9751
	LSTM	3.3407E-05	3.8521E-09	0.4884	0.8978
	GRU	3.3188E-05	3.8436E-09	0.4800	0.8122
	DBN-LSTM-GRU	3.2772E-05	3.5182E-09	0.4944	0.8664
	GVMD-DBN-LSTM-GRU	2.4617E-05	9.0556E-10	0.4715	0.3211
	Q-DBN-LSTM-GRU	2.7079E-05	2.0183E-09	0.4708	0.3977
	WT-Q-DBN-LSTM-GRU	2.3504E-05	1.8798E-09	0.3166	0.1997
	GVMD-Q-DBN-LSTM-GRU	1.4863E-05	6.8613E-10	0.2228	0.1214
SPX	BPNN	6.1348E-05	3.3765E-08	0.7914	1.4135
	ELM	6.2175E-05	3.3787E-08	0.7990	1.4615
	SVR	6.1007E-05	4.3988E-08	0.7576	1.3040
	DBN	5.7288E-05	3.1096E-08	0.7282	1.1709
	LSTM	5.3436E-05	2.9557E-08	0.6858	0.9798
	GRU	5.5428E-05	3.1050E-08	0.6817	0.9855
	DBN-LSTM-GRU	5.4258E-05	2.9781E-08	0.6876	0.9865
	GVMD-DBN-LSTM-GRU	4.0858E-05	1.4733E-08	0.5691	0.8970
	Q-DBN-LSTM-GRU	4.3118E-05	1.7910E-08	0.6392	0.7701
	WT-Q-DBN-LSTM-GRU	3.8492E-05	1.3682E-08	0.4515	0.3667
	GVMD-Q-DBN-LSTM-GRU	2.5400E-05	5.3357E-09	0.3698	0.3606
FTSE	BPNN	6.9014E-05	6.0040E-08	0.7882	2.5096
	ELM	6.8930E-05	5.8872E-08	0.9190	3.4646
	SVR	6.7266E-05	6.2676E-08	0.9142	3.4276
	DBN	6.6950E-05	6.1476E-08	0.6139	1.1645
	LSTM	6.4334E-05	5.6812E-08	0.5999	0.9292
	GRU	6.3013E-05	5.7391E-08	0.5586	0.8817
	DBN-LSTM-GRU	6.3392E-05	5.7549E-08	0.5727	0.9360
	GVMD-DBN-LSTM-GRU	5.1435E-05	3.9355E-08	0.4387	0.7448
	Q-DBN-LSTM-GRU	5.5265E-05	4.8511E-08	0.5206	0.4687
	WT-Q-DBN-LSTM-GRU	5.4943E-05	3.6294E-08	0.5036	0.4646
	GVMD-Q-DBN-LSTM-GRU	4.6555E-05	3.5065E-08	0.3820	0.2981

Note: Bolded numbers indicate the lowest value of loss function.

3.5. Comparison and analysis

In the cause of obtaining the primarily baseline model for RV sequence of three datasets SSEC, SPX and FTSE, the representative models in a single model, including BPNN, ELM, SVR, DBN, LSTM and GRU are selected for comparison. On the basis of the evaluating indicators mentioned above, the performance of forecasting model is evaluated by MAE, MSE, HMAE and HMSE. The following conclusions can be clearly concluded from Fig. 9 and Table 3.

The estimation errors of DBN, LSTM and GRU, which are deep neural networks, are less than those of the above methods, and the forecasting outcomes on the same dataset are not much different. The reason is that the deep learning model has more hidden layers, so it can fully capture the nonlinear characteristics of RV sequence. However, DBN does not perform excellent under all evaluation metrics in FTSE dataset compared with traditional prediction models. Although the MAE, HMAE and HMSE of DBN (6.6950E-05, 0.6139, 1.1645) are smaller than BPNN (6.9014E-05, 0.7882, 2.5096), ELM (6.8930E-05, 0.9190, 3.4646) and SVR (6.7266E-05, 0.9142, 3.4276), the MSE of DBN is 6.1476E-08, which is larger than the above three models (6.0040E-08, 5.8872E-08, 6.2676E-08). The reason is that MSE is sensitive to the extreme values in the dataset, resulting in a larger MSE value.

Furthermore, it should be pointed out that owing to the discrepancies in the structure of deep neural network models, their capability to recognize variant datasets is divergent. For instance, in the dataset of SPX, the performance of LSTM (MAE=5.3436E-05, MSE=2.9557E-08, HMAE=0.6858, HMSE=0.9798) and GRU (MAE=5.5428E-05, MSE=3.1050E-08, HMAE=0.6817, HMSE=0.9855) is better than that of DBN (MAE=5.7288E-05, MSE=3.1096E-08, HMAE=0.7282, HMSE=1.1709), while DBN (MSE=3.3250E-09) show the excellent prediction performance than LSTM and GRU (MSE=3.8521E-09 and 3.8436E-09) on the dataset of SSEC, which indicates that it is necessary to adopt an integration strategy to improve the generalization and robustness of the prediction model.

In order to prove that the Q-learning algorithm has advantages in improving the prediction ability of the integrated model, this paper compares the integrated model Q-DBN-LSTM-GRU with DBN, LSTM and GRU method and DBN-LSTM-GRU. The DBN-LSTM-GRU model indicates that the weights of three baseline models are set to 1/3, which means the role of Q-learning algorithm in weight optimization is eliminated. Table 4 lists the performance improvement percentage of Q-DBN-LSTM-GRU on the baseline model.

As can be seen from Table 3 and Table 4, the forecasting accuracy of Q-DBN-LSTM-GRU is higher than that of DBN, LSTM and GRU in all three stock markets. Taking the SSEC dataset as an example, the Q-DBN-

Table 4

The improvement percentages of the Q-DBN-LSTM-GRU model for the baseline models.

Dataset	Model	$P_{MAE}(\%)$	$P_{MSE}(\%)$	$P_{HMAE}(\%)$	$P_{HMSE}(\%)$
SSEC	DBN vs.Q-DBN-LSTM-GRU	18.8377	39.2992	14.7565	59.2144
	LSTM vs.Q-DBN-LSTM-GRU	18.9421	47.6052	3.6036	55.7028
	GRU vs.Q-DBN-LSTM-GRU	18.4073	47.4893	1.9167	51.0342
SPX	DBN vs.Q-DBN-LSTM-GRU	24.7347	42.4042	12.2219	34.2301
	LSTM vs.Q-DBN-LSTM-GRU	19.3091	39.4052	6.7950	21.4023
	GRU vs.Q-DBN-LSTM-GRU	22.2090	42.3188	6.2344	21.8569
FTSE	DBN vs.Q-DBN-LSTM-GRU	17.4533	21.0895	15.1979	59.7510
	LSTM vs.Q-DBN-LSTM-GRU	14.0967	14.6113	13.2189	49.5588
	GRU vs.Q-DBN-LSTM-GRU	12.2959	15.4728	6.8027	46.8413

LSTM-GRU with MAE, MSE, HMAE and HMSE are 2.7079E-05, 2.0183E-09, 0.4708 and 0.3977, respectively, which further improved the prediction accuracy compared with DBN, LSTM and GRU. The MAE improvement percentages of the Q-DBN-LSTM-GRU integrated by Q-learning algorithm compared with baseline model DBN, LSTM and GRU are 18.8377%, 18.9421% and 18.4073%, respectively. Without doubt, integrated models can also obtain high accuracy prediction results on other datasets with stronger generalization. In addition, the Q-learning algorithm can be used to adaptively adjust the weight according to the characteristics of the dataset, complementing the advantages and disadvantages of each baseline model. For example, the MSE improvement percentages of Q-DBN-LSTM-GRU compared with DBN in three datasets SSEC, SPX and FTSE are 39.2992%, 42.4042% and 21.0895%, respectively, which improves the overall prediction accuracy of the integrated model.

In addition, compared with the Q-DBN-LSTM-GRU model in the SSEC dataset, the prediction error of DBN-LSTM-GRU model is larger, and MAE, MSE, HMAE and HMSE are higher, which are 3.2772E-05, 3.5182E-09, 0.4944 and 0.8664, respectively. At the same time, it also can be seen that the DBN-LSTM-GRU model is superior to DBN, LSTM and GRU on MAE (3.3364E-05, 3.3407E-05, 3.3188E-05), but is higher than the values of DBN (3.3250E-09) and lower than the values of LSTM (3.8521E-09) and GRU (3.8436E-09) in MSE, higher than the values of LSTM (0.4884) and GRU (0.4800) and lower than the values of DBN (0.5523) in HMAE, higher than the values GRU (0.8122) and lower than the values of DBN (0.9751) and LSTM (0.8978) in HMSE. This further demonstrates the effectiveness of Q-learning algorithm in optimizing the baseline model weights.

With the aim of comparing and screening decomposition methods with better feature recognition capability to improve the forecasting power of the ensemble model, this study compares the prediction outcomes of DBN-LSTM-GRU model with GVMD-DBN-LSTM-GRU model, Q-DBN-LSTM-GRU model with GVMD-Q-DBN-LSTM-GRU and WT-Q-DBN-LSTM-GRU model, and the results are presented in Table 3 and Table 5.

Taking the SSEC dataset as an example, the prediction performance of DBN-LSTM-GRU and Q-DBN-LSTM-GRU model are significantly

Table 5

The improvement percentages of the ensemble models for the different decomposition methods.

Dataset	Model	$P_{MAE}(\%)$	$P_{MSE}(\%)$	$P_{HMAE}(\%)$	$P_{HMSE}(\%)$
SSEC	DBN-LSTM-GRU vs. GVMD-DBN-LSTM-GRU	24.8840	74.2607	4.6319	62.9386
	Q-DBN-LSTM-GRU vs. WT-Q-DBN-LSTM-GRU	13.2021	6.8622	32.7528	49.7863
	Q-DBN-LSTM-GRU vs. GVMD-Q-DBN-LSTM-GRU	45.1124	66.0046	52.6763	69.4745
	Q-DBN-LSTM-GRU vs. WT-Q-DBN-LSTM-GRU	10.7287	23.6069	29.3648	52.3828
SPX	DBN-LSTM-GRU vs. GVMD-DBN-LSTM-GRU	24.6968	50.5289	17.2339	9.0725
	Q-DBN-LSTM-GRU vs. WT-Q-DBN-LSTM-GRU	41.0919	70.2083	42.1464	53.1749
	Q-DBN-LSTM-GRU vs. GVMD-Q-DBN-LSTM-GRU	18.8620	31.6148	23.3979	20.4274
	Q-DBN-LSTM-GRU vs. WT-Q-DBN-LSTM-GRU	0.5826	25.1840	3.2655	0.8748
FTSE	DBN-LSTM-GRU vs. GVMD-DBN-LSTM-GRU	15.7604	27.7174	26.6231	36.3985
	Q-DBN-LSTM-GRU vs. WT-Q-DBN-LSTM-GRU				
	Q-DBN-LSTM-GRU vs. GVMD-Q-DBN-LSTM-GRU				
	Q-DBN-LSTM-GRU vs. WT-Q-DBN-LSTM-GRU				

enhanced after preprocessing of RV sequence. The predictive error of the GVMD-DBN-LSTM-GRU model is smaller than DBN-LSTM-GRU model, and the errors of both WT-Q-DBN-LSTM-GRU and GVMD-Q-DBN-LSTM-GRU model are smaller than Q-DBN-LSTM-GRU model. The original RV sequence has obvious volatility, which brings great challenges to modeling. Fortunately, the decomposition method decomposes the original sequence into IMFs with different economic implications, which facilitates the neural network to capture the features contained in the sequence more efficiently and enhance the model predictive performance. Among all the decomposition methods, VMD is consistently the best. As can be obtained from Table 5, the P_{MAE} , P_{MSE} , P_{HMAE} and P_{HMSE} of WT-Q-DBN-LSTM-GRU relative to Q-DBN-LSTM-GRU are 13.2021%, 6.8622%, 32.7528% and 49.7863%, respectively. The P_{MAE} , P_{MSE} , P_{HMAE} and P_{HMSE} of GVMD-Q-DBN-LSTM-GRU relative to Q-DBN-LSTM-GRU are 45.1124%, 66.0046%, 52.6763% and 69.4745%, respectively.

Based on the above experimental results, the VMD technique can decompose the RV sequence into subsequences with diverse frequencies, and the GWO algorithm can effectively reduce the randomness of VMD parameter selection, thereby significantly improving the forecasting accuracy of the model.

In the above analysis, although the forecasting precision of the model

Table 6

The MDM results of GVMD-Q-DBN-LSTM-GRU compared with other reference models.

Dataset	Model	MAE	MSE	MDM1	MDM2
SSEC	BPNN	0.0000***	0.0000***	-20.2205	-7.1352
	ELM	0.0000***	0.0000***	-18.2921	-7.1095
	SVR	0.0000***	0.0000***	-17.6284	-7.0626
	DBN	0.0000***	0.0000***	-16.9309	-7.6156
	LSTM	0.0000***	0.0000***	-15.6476	-6.8220
	GRU	0.0000***	0.0000***	-15.3330	-6.9082
	DBN-LSTM-GRU	0.0000***	0.0000***	-15.8439	-7.1912
	GVMD-DBN-LSTM-GRU	0.0000***	0.0012***	-15.7683	-3.2513
	Q-DBN-LSTM-GRU	0.0000***	0.0000***	-13.1563	-4.5772
	WT-Q-DBN-LSTM-GRU	0.0000***	0.0000***	-10.7912	-5.0749
SPX	BPNN	0.0000***	0.0073***	-10.0176	-2.6884
	ELM	0.0000***	0.0079***	-10.3368	-2.6602
	SVR	0.0000***	0.0084***	-7.8647	-2.6418
	DBN	0.0000***	0.0120**	-9.4116	-2.5151
	LSTM	0.0000***	0.0183**	-8.6301	-2.3638
	GRU	0.0000***	0.0131**	-8.7564	-2.4843
	DBN-LSTM-GRU	0.0000***	0.0236**	-8.5340	-2.2661
	GVMD-DBN-LSTM-GRU	0.0000***	0.0013***	-7.5243	-3.2137
	Q-DBN-LSTM-GRU	0.0000***	0.0671*	-5.5334	-1.8325
	WT-Q-DBN-LSTM-GRU	0.0000***	0.0085***	-6.5353	-2.6346
FTSE	BPNN	0.0000***	0.0677*	-7.2765	-1.4932
	ELM	0.0000***	0.0791*	-6.6869	-1.4110
	SVR	0.0000***	0.0554*	-8.2407	-1.5949
	DBN	0.0000***	0.0592*	-7.3130	-1.5618
	LSTM	0.0000***	0.0910*	-6.9746	-1.3346
	GRU	0.0000***	0.0900*	-6.4618	-1.3410
	DBN-LSTM-GRU	0.0000***	0.0874*	-6.7318	-1.3572
	GVMD-DBN-LSTM-GRU	0.0011***	0.0583*	-3.2717	-1.5689
	Q-DBN-LSTM-GRU	0.0021***	0.0298**	-3.0855	-1.8833
	WT-Q-DBN-LSTM-GRU	0.0000***	0.2912	-3.8928	-0.5500

Note: *, ** and *** denote rejection of the null hypothesis at the 10%, 5% and 1% significance level respectively. MDM1 and MDM2 represent the values of MDM test statistics based on MAE and MSE loss functions, respectively. H_0 : GVMD-Q-DBN-LSTM-GRU model has the same predictive performance as other contrast models. H_1 : The predictive power of the GVMD-Q-DBN-LSTM-GRU model is different from other contrast models.

is compared ground on the four evaluation indicators MAE, MSE, HMAE and HMSE and their corresponding improvement percentages, it does not determine whether there is a significant difference in the prediction effect between them. Therefore, we use GVMD-Q-DBN-LSTM-GRU as the target model and BPNN, ELM, SVR, DBN, LSTM, GRU, DBN-LSTM-GRU, GVMD-DBN-LSTM-GRU, Q-DBN-LSTM-GRU and WT-Q-DBN-LSTM-GRU as the contrast model for MDM test. In our study, MAE and MSE are applied as the loss function, and the MDM test result values are listed in Table 6.

It can be intuitively seen from Table 6 that for the SSEC dataset, there are all cases rejected the null hypothesis at the 1% significance level. For the SPX dataset, there are 15 cases rejected the null hypothesis at the 1% significance level, 4 cases rejected the null hypothesis at the 5% significance level, and 1 case rejected the null hypothesis at the 10% significance level. For the FTSE dataset, there are 10 cases rejected the null hypothesis at the 1% significance level, 1 case rejected the null hypothesis at the 5% significance level, and 8 cases rejected the null hypothesis at the 10% significance level. The above analysis indicates that the difference between the GVMD-Q-DBN-LSTM-GRU model proposed in our paper and other comparative models are statistically significant, and the prediction ability of GVMD-Q-DBN-LSTM-GRU model exceeds that of other models.

In summary, the superiority of GVMD-Q-DBN-LSTM-GRU model still exists, and its forecast behavior is better than other reference models.

3.6. Robustness check

Investors pay attention not only to the short-term prediction performance of the model, but also on the medium-term and long-term

Table 8

Hyperparameter settings selection of baseline models at two other horizons. (H = 5/22).

Model	Name of Parameter	SSEC	SPX	FTSE
DBN	Number of the Hidden units	[25,10]/ [25,10]	[50,20]/ [25,10]	[25,10]/ [25,10]
	Learning rate in Pre-training process	0.001/0.001	0.0001/ 0.001	0.0001/ 0.0001
	Learning rate in Fine-tuning process	0.0001	0.0001	0.0001
	Number of the Hidden units	256/128	32/16	32/32
LSTM	Optimizer	Adam	Adam	Adam
	Dropout	0.2/0.2	0.2/0.3	0.2/0.2
	Number of the Hidden units	128/128	32/32	16/16
GRU	Optimizer	Adam	Adam	Adam
	Dropout	0.2/0.3	0.3/0.3	0.3/0.3

prediction ability of the model. Therefore, our study further measures the prediction accuracy of the basic model and the extended model in 5 days and 22 days through robustness check.

We mainly did two aspects of work. Firstly, the multi-step prediction accuracy of the GVMD-Q-DBN-LSTM-GRU model and the comparison model under four evaluation indexes of MAE, MSE, HMAE and HMSE is compared, and the specific results are displayed in Table 7. It is not difficult to find that the model proposed in this paper has the most prominent ability to predict the RV series of stock price indices, and the hyperparameter settings of the baseline model are shown in Table 8. Taking the 5-step and 22-step forecasting of SSEC as an example, the

Table 7

The deterministic forecasting results of the proposed models and compared models at two other horizons. (H = 5/22).

Model	H = 5				H = 22			
	MAE	MSE	HMAE	HMSE	MAE	MSE	HMAE	HMSE
SSEC								
BPNN	5.8770E-05	6.7101E-09	1.2864	4.1124	7.1623E-05	8.2127E-09	1.7339	6.6983
ELM	4.8639E-05	6.1635E-09	0.9208	2.7645	6.1573E-05	7.4050E-09	1.3687	4.6503
SVR	5.3498E-05	5.7708E-09	1.1987	3.3727	6.1846E-05	6.8079E-09	1.4678	4.8307
DBN	4.4369E-05	5.9130E-09	0.7577	1.9823	5.5892E-05	7.1038E-09	1.1289	3.3656
LSTM	4.3976E-05	5.9800E-09	0.6634	1.7121	5.6673E-05	7.2762E-09	1.1377	3.5772
GRU	4.3692E-05	6.0453E-09	0.6393	1.5955	5.9685E-05	7.1669E-09	1.2925	4.1014
DBN-LSTM-GRU	4.3297E-05	5.9224E-09	0.6653	1.7136	5.6904E-05	7.0418E-09	1.1806	3.5961
GVMD-DBN-LSTM-GRU	2.8927E-05	1.9549E-09	0.6076	0.7766	5.3096E-05	5.7613E-09	1.0706	1.9462
Q-DBN-LSTM-GRU	3.8642E-05	4.7819E-09	0.5715	0.6025	4.8680E-05	5.7049E-09	0.9120	2.4115
WT-Q-DBN-LSTM-GRU	3.0425E-05	2.2240E-09	0.5085	0.4125	4.6715E-05	5.5464E-09	0.8613	1.6392
GVMD-Q-DBN-LSTM-GRU	2.2408E-05	1.6202E-09	0.3199	0.2486	3.8927E-05	5.5324E-09	0.5478	0.5717
SPX								
BPNN	8.6833E-05	6.0933E-08	1.5542	6.1797	1.0915E-04	1.0330E-07	2.0754	12.0880
ELM	8.7123E-05	6.2171E-08	1.3928	5.1288	1.1188E-04	1.0469E-07	2.2558	13.1130
SVR	7.5630E-05	5.8762E-08	1.2921	4.2724	1.0884E-04	9.6497E-08	2.0680	12.0858
DBN	7.4209E-05	6.1839E-08	1.0749	2.7518	1.0757E-04	9.7435E-08	2.1771	12.3453
LSTM	7.4801E-05	5.7283E-08	0.9345	2.2167	1.0239E-04	9.5682E-08	2.0379	11.2157
GRU	7.6449E-05	5.5918E-08	1.0421	2.8104	1.0207E-04	9.6324E-08	1.9777	10.6998
DBN-LSTM-GRU	7.1791E-05	5.4468E-08	0.9696	2.1859	1.0346E-04	9.6102E-08	2.0595	11.2460
GVMD-DBN-LSTM-GRU	4.8553E-05	2.0084E-08	0.6391	1.5415	9.2775E-05	6.7992E-08	1.8255	8.2612
Q-DBN-LSTM-GRU	6.0102E-05	3.2813E-08	0.8300	1.2827	9.3550E-05	8.8476E-08	1.3956	5.1470
WT-Q-DBN-LSTM-GRU	5.6610E-05	2.7409E-08	0.7071	1.0116	8.7258E-05	8.8147E-08	1.0594	3.5425
GVMD-Q-DBN-LSTM-GRU	3.5036E-05	8.6934E-09	0.5650	1.0084	7.4315E-05	4.5017E-08	1.0318	2.5983
FTSE								
BPNN	7.9684E-05	7.2708E-08	1.1153	5.1655	1.0438E-04	1.0344E-07	1.6881	10.5783
ELM	8.1051E-05	7.1476E-08	1.2333	5.9808	1.0324E-04	1.0466E-07	1.5775	8.8475
SVR	7.6596E-05	7.4829E-08	1.1124	5.1163	1.0426E-04	1.0346E-07	1.6915	11.0507
DBN	7.7718E-05	8.0552E-08	1.0822	4.8378	9.7070E-05	1.0386E-07	1.3073	6.1174
LSTM	7.0639E-05	6.9936E-08	0.8589	3.0066	9.1609E-05	1.0410E-07	1.0611	4.3138
GRU	6.9895E-05	6.8712E-08	0.8057	2.6556	8.9769E-05	1.0435E-07	1.0054	4.1902
DBN-LSTM-GRU	7.0647E-05	7.1638E-08	0.8981	3.3285	9.1174E-05	1.0316E-07	1.1110	4.7377
GVMD-DBN-LSTM-GRU	5.9882E-05	4.5545E-08	0.5705	1.1680	8.9128E-05	9.9719E-08	0.9680	1.9636
Q-DBN-LSTM-GRU	5.8977E-05	5.9392E-08	0.7154	1.0951	8.6326E-05	9.8639E-08	0.9756	3.9818
WT-Q-DBN-LSTM-GRU	5.3606E-05	4.3302E-08	0.6122	0.6019	8.4567E-05	9.1570E-08	0.9695	2.2873
GVMD-Q-DBN-LSTM-GRU	5.2893E-05	3.5198E-08	0.4722	0.3771	8.2165E-05	8.9858E-08	0.7569	1.0592

Note: Bolded numbers indicate the lowest value of loss function.

Table 9

The MDM results of GVMD-Q-DBN-LSTM-GRU compared with other reference models at two other horizons. (H = 5/22).

Model	H = 5				H = 22			
	MAE	MSE	MDM1	MDM2	MAE	MSE	MDM1	MDM2
SSEC								
BPNN	0.0000***	0.0000***	-30.2719	-10.6868	0.0000***	0.0000***	-22.0593	-8.8469
ELM	0.0000***	0.0000***	-21.4248	-8.6008	0.0000***	0.0000***	-17.5799	-7.1550
SVR	0.0000***	0.0000***	-29.8845	-8.7077	0.0000***	0.0000***	-19.2357	-5.6406
DBN	0.0000***	0.0000***	-18.4891	-7.7597	0.0000***	0.0000***	-14.2579	-6.2632
LSTM	0.0000***	0.0000***	-18.0924	-7.6230	0.0000***	0.0000***	-13.6224	-5.9332
GRU	0.0000***	0.0000***	-17.7475	-7.5397	0.0000***	0.0000***	-15.6493	-5.6046
DBN-LSTM-GRU	0.0000***	0.0000***	-16.9979	-7.2175	0.0000***	0.0000***	-12.8284	-4.9591
GVMD-DBN-LSTM-GRU	0.0000***	0.0000***	-12.0848	-3.9046	0.0000***	0.4985	-13.0039	-0.6771
Q-DBN-LSTM-GRU	0.0000***	0.0000***	-14.8754	-6.5330	0.0000***	0.5278	-8.1411	-0.6316
WT-Q-DBN-LSTM-GRU	0.0000***	0.0000***	-11.4000	-4.8323	0.0000***	0.9649	-7.0124	-0.0441
SPX								
BPNN	0.0000***	0.0006***	-10.4525	-3.4423	0.0000***	0.0000***	-6.9482	-4.1002
ELM	0.0000***	0.0002***	-10.1425	-3.7132	0.0000***	0.0000***	-7.2135	-4.1073
SVR	0.0000***	0.0038***	-8.2194	-2.8988	0.0000***	0.0002***	-7.1210	-3.7930
DBN	0.0000***	0.0005***	-10.7580	-3.4925	0.0000***	0.0001***	-7.0659	-3.8143
LSTM	0.0000***	0.0020***	-8.1859	-3.0969	0.0000***	0.0002***	-6.0709	-3.6946
GRU	0.0000***	0.0013***	-8.6511	-3.2267	0.0000***	0.0002***	-5.9362	-3.7159
DBN-LSTM-GRU	0.0000***	0.0004***	-10.1019	-3.5486	0.0000***	0.0002***	-6.2738	-3.7173
GVMD-DBN-LSTM-GRU	0.0000***	0.0040***	-7.1770	-2.8804	0.0000***	0.0012***	-6.0275	-3.2464
Q-DBN-LSTM-GRU	0.0000***	0.0457**	-6.3324	-2.0000	0.0000***	0.0008***	-4.7881	-3.3751
WT-Q-DBN-LSTM-GRU	0.0000***	0.0016***	-7.6818	-3.1680	0.0012***	0.0013***	-3.2517	-3.2340
FTSE								
BPNN	0.0000***	0.0118**	-7.4362	-2.2622	0.0000***	0.0000***	-11.0374	-4.4129
ELM	0.0000***	0.0140**	-7.8971	-2.1984	0.0000***	0.0000***	-10.3730	-4.6238
SVR	0.0000***	0.0166**	-6.2523	-2.1302	0.0000***	0.0000***	-9.2037	-4.2370
DBN	0.0000***	0.0120**	-5.9885	-2.2572	0.0000***	0.0000***	-6.9694	-4.2660
LSTM	0.0000***	0.0220**	-5.0175	-2.0138	0.0000***	0.0000***	-5.4941	-4.2985
GRU	0.0000***	0.0228**	-4.9744	-1.9989	0.0000***	0.0000***	-4.4798	-4.2787
DBN-LSTM-GRU	0.0000***	0.0204**	-4.8730	-2.0462	0.0000***	0.0000***	-5.2848	-4.0741
GVMD-DBN-LSTM-GRU	0.0002***	0.0221**	-3.7264	-2.0131	0.0000***	0.0005***	-6.2412	-3.3184
Q-DBN-LSTM-GRU	0.1107	0.0789*	-1.5962	-1.4122	0.0342**	0.0024***	-2.1200	-2.8237
WT-Q-DBN-LSTM-GRU	0.8350	0.1263	-0.2083	-1.1439	0.4051	0.3323	-0.8328	-0.4335

Note: *, ** and *** denote rejection of the null hypothesis at the 10%, 5% and 1% significance level respectively. MDM1 and MDM2 represent the values of MDM test statistics based on MAE and MSE loss functions, respectively. H_0 : GVMD-Q-DBN-LSTM-GRU model has the same predictive performance as other contrast models. H_1 : The predictive power of the GVMD-Q-DBN-LSTM-GRU model is different from other contrast models.

evaluating indexes of GVMD-Q-DBN-LSTM-GRU are (2.2408E-05, 1.6202E-09, 0.3199, 0.2486) and (3.8927E-05, 5.5324E-09, 0.5478, 0.5717), respectively, which are smaller than the rest of comparison models, and our presented method also demonstrates the superior behavior when predicting the RV series of the remaining two datasets.

Secondly, the MDM test is used to verify whether the proposed model significantly outperforms the comparison model, and the test results in Table 9 strengthen the above conclusions. In the 5-step and 22-step prediction of SSEC, SPX and FTSE, the MDM test results are significant in the vast majority of cases, indicating that our proposed GVMD-Q-DBN-LSTM-GRU model still outperformed other models, which also confirms the strong generalization of GVMD-Q-DBN-LSTM-GRU.

4. Conclusion

The prediction of RV series is a key step in risk management, portfolio allocation and derivative pricing, which is of great significance to stabilize the financial market order. Based on this, a novel forecasting model of RV series GVMD-Q-DBN-LSTM-GRU is proposed, which introduces decomposition method, deep learning and reinforcement learning technology. First of all, the VMD optimized by GWO algorithm decomposes the RV series into IMFs of different frequencies. Secondly, employing the deep learning models DBN, LSTM and GRU to predict the identical IMF concurrently. Finally, the Q-learning algorithm is adopted to integrate the forecasting results of DBN, LSTM and GRU to obtain the final results. In addition, we take three representative stock price indices SSEC, SPX and FTSE in the financial market as empirical samples. On the ground of evaluation indexes MAE, MSE, HMAE and HMSE, we compare

the prediction accuracy of BPNN, ELM, SVR, DBN, LSTM, GRU, DBN-LSTM-GRU, GVMD-DBN-LSTM-GRU, Q-DBN-LSTM-GRU and WT-Q-DBN-LSTM-GRU with the proposed GVMD-Q-DBN-LSTM-GRU model, and further verify the superiority of the presented model by MDM test.

After repeated experimental verification, the results show that:

- The GVMD-Q-DBN-LSTM-GRU model sufficiently combines outstanding decomposition method, reinforcement learning integration technique and deep neural networks. That is why the model proposed in our paper has better prediction ability and adaptability in the RV sequence of three stock price indices, and outperforms all comparative models in the experiment.
- Deep neural network has more hidden layers, which contribute to fully capture the nonlinear characteristics of RV series. For this reason, the estimation errors of DBN, LSTM and GRU are smaller than those of traditional prediction methods, and the prediction results on the same dataset are not much different, which becomes the baseline model for the integration of Q-learning algorithm.
- The Q-learning algorithm in reinforcement learning realizes the adaptive optimization of the weight coefficient of the integrated model through the interaction between the agent and the environment, and has great algorithm analysis ability. Therefore, the Q-learning algorithm can effectively integrate three deep neural networks, and the Q-DBN-LSTM-GRU model has better prediction capability than single models DBN, LSTM, GRU and DBN-LSTM-GRU model with equally weight distribution.

- (d) VMD is an adaptive non-recursive sequence decomposition technique, which can effectively improve the accuracy of the prediction method. The GWO algorithm can avoid errors caused by artificial setting the VMD parameters due to the lack of prior knowledge. As a consequence, using more appropriate parameters can improve the decomposition efficiency, and GVMD-Q-DBN-LSTM-GRU and WT-Q-DBN-LSTM-GRU model with decomposition technology have higher prediction efficiency than Q-DBN-LSTM-GRU model, and also, the forecasting precision of GVMD-DBN-LSTM-GRU model is also higher than DBN-LSTM-GRU model. However, the capability of VMD method for feature extraction of RV sequences is better than WT method.
- (e) In the robustness check, we use longer forecasting steps (5-step forecasting and 22-step forecasting) to prove that the conclusion is not a coincidence. The MDM test results further verify the effectiveness of GVMD-Q-DBN-LSTM-GRU model for the RV series prediction of stock price index.

In a word, the performance of the ensemble model proposed in this paper can provide more accurate results in 1-step, 5-step and 22-step forecasting, which is an ideal RV series prediction model. It is worth noting that our study has important theoretical value and practical significance for investors to adjust trading decisions and avoid risks in a timely manner to maximize profits, as well as for regulators to discover abnormal trends, and ultimately promote the steady and rapid development of financial markets.

Conclusively, the limitation of this study and direction for future research are pointed out. The limitation of this paper is that multivariate prediction is not considered. In future research work, we will consider other factors closely related to the RV series of stock price index for RV series forecasting.

CRedit authorship contribution statement

Yuanyuan Yu: Conceptualization, Methodology, Software, Writing – original draft. **Yu Lin:** Data curation, Validation, Supervision, Writing – review & editing. **Xianping Hou:** Software, Visualization, Validation. **Xi Zhang:** Formal analysis, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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